



The Power of Flight



LEAP-1B

Maintenance Guide

Oct 2016 , Rev 00

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LEAP



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REVISION STATUS

NOTE: The modifications introduced in this LEAP MAINTENANCE GUIDE revision are highlighted in yellow.

Revision Date	Summary of changes

All technical documentation and information contained herein, with respect to assembly and disassembly, cleaning, inspection methods and limits, repair methods and limits, operational limits, life limits and the like, have been developed and approved for use with engines and parts that have been manufactured and/or approved by CFM and that have been maintained in accordance with CFM technical documentation and recommendations.

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Section 1: General Maintenance Information



Introduction

This document provides maintenance guidelines for off-wing maintenance for the LEAP 1B engine model.

This document is to be used in conjunction with agency approved engine manufacturer's manuals and service bulletins. If any conflict arises between this document and the authorized manual material, the authorized manual material and applicable service bulletins are considered the controlling authority.

Purpose

The intention of this document is to provide to Lessors and Airlines an help to define proper off-wing LEAP engine maintenance in order to achieve engine reliability, performance, and extended time on wing in a cost effective manner.

This document will be issued and updated by CFM PSE.

Manual References

Aircraft Maintenance Manual
LEAP 1B Engine Shop Manual
CFM Standard Practice Manual
CFM Consumable Products Manual
LEAP Illustrated Tools and Equipment Manual
CFM Non Destructive Test Manual

Abbreviations

AD Airworthiness Directive
AIPC Aircraft Illustrated Part Catalog
AMM Aircraft Maintenance Manual
ATA Airline Transport Association
BFE Buyer Furnished Equipment
BSI Borescope Inspection
C&A Controls and Accessories
CDP Compressor Discharge Pressure
CDR Customer Departure Record
CMM Component Maintenance Manual
CSLSV Cycles Since Last Shop Visit
CSN Cycles Since New



CSO Cycles Since Overhaul
CTWS Continued Time Workscope
DOD Domestic Object Damage (Damage originating as a result of an internal engine failure)
EASA European Aviation Safety Agency
EDS Engine Data Submittal
EGT Exhaust Gas Temperature
ESM Engine Shop Manual
FAA Federal Aviation Administration
FOD Foreign Object Damage (Damage from material originating outside engine)
CC Customer Connect
GVI General Visual Inspection
HSI Hot Section Intervention
LLP Life Limited Part, as defined in Engine Manual Chapter 5
LRU Line Replaceable Unit
LEAP MG LEAP Maintenance Guide
MPD Maintenance Planning Document
OC On Condition
PCA Part Condition Assessment
PSE Product Support Engineering
QEC Quick Engine Change
Q/T Quick Turn
SB Service Bulletin
SE Services Engineering
SM Shop Module (= minor module)
SP Special Procedure
SV Shop Visit
TOW Time On Wing
TSLSV Time Since Last Shop Visit
TSN Time Since New
TSO Time Since Overhaul, applies to Modules only, not Engine-level worksopes



Definitions

Airworthiness Limitations

LEAP-1B Airworthiness Limitations are defined in ESM ATA sections 05-11/12/21 (Life Limits) and 05-21 (Mandatory Inspections). The Airworthiness Limitations are FAA and EASA approved and must be complied with as part of the engine maintenance action.

Engine Build Standard

The Engine Build Standard is the TOW (number of hours and cycles) that the shop visit is anticipated to yield based on an assessment of factors that contribute to engine removal (LLP life, module workscope, EGT deterioration rate, etc.).

Engine Data Submittal

All new production engines are delivered with an Engine Data Submittal (EDS). This document provided by the Customer Web Center, is used to establish the configuration data for the delivered engine.

Engine Exceedance

An engine exceedance is a non-standard Operational event or an engine parameter that has gone beyond operating limits. An engine that has experienced an exceedance must have a special workscope defined, typically with the guidance of the CFM SE/PSE teams. The following are examples of Operational events that could lead to an engine exceedance:

- Stall; FOD/DOD; Low or Loss of Oil Pressure; N1 or N2 Rotor Overspeed
- Engine Overtemperature; EGT Exceedance; Volcanic Ash Ingestion; High N1/N2 Vibration
- Metal Contamination of Oil System; Exposure to Fire or Extinguishing Agents

Engine Minimum Stub Life (also known as "Minimum Build")

The minimum engine stub life for a proposed engine shop visit, expressed in "cycles remaining".

It should be noted that in the case of thrust levels affecting maximum LLP life, the thrust level should also be referenced when specifying the Engine Minimum Stub Life.

Hard Time

Hard time limits, such as Life Limited Parts (LLPs), which are controlled through Chapter 05-11/12-XX in the LEAP 1B Engine Manual. Chapter 05-11/12/21-XX the "Airworthiness Limitation" section for life limited parts requires the adoption of an Airworthiness Directive (AD) to mandate the change.

Stub Life

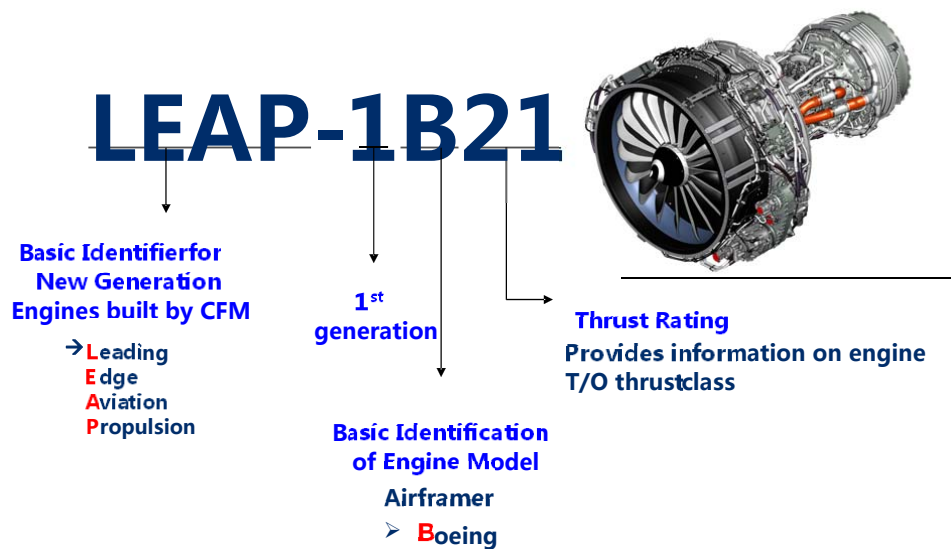
The Stub life of a component is the cycles remaining below the mandatory retirement life limit on a Life Limited Part (LLP) for the intended installation application. The life limit is often specific to the thrust or model application. The "stub-life" of the engine is the shortest life remaining of all life-limited parts installed in the engine. It is desirable to plan the stub-life such that engines are rarely or never removed for part life expiration while not limiting Time On Wing (TOW). Minimum stub life for a module or unit corresponds to the lowest remaining part of all the life limited parts in a module or unit, respectively.

**Soft time**

Soft time limits are recommended limits, for hardware, above which specified levels of maintenance should be performed to ensure the hardware will perform within the applicable maintenance period for engine shop visits other than convenience removals.

Life Limits

Chapter 5-11/12/21 defines the specific Rotating and Static Structure parts that are life limited with regard to time in service. The Life Limits are specified in terms of Flight cycles.

LEAP Name**ESN sequence update**

The LEAP 1B ESN will be assigned as 602- then 603- then ... 60A.

Examples : 602-101 to 602-999 and 603-101 to 603-999.



Mandatory Inspections

Chapter 5-21 of the LEAP-1B Series Engine Manual defines several Mandatory inspections for the engine Life Limited Parts. These inspections shall be applied when the LLP is exposed at piece part during an overhaul workscope. Inspections may also be applicable at module level if specified in the module level inspection.

Maintenance Philosophy

On-Condition (OC)

The LEAP engine family utilizes an OC Maintenance philosophy. This approach allows flexibility in defining the engine level overhaul workscope based on an assessment of the engine condition from on-wing data, LLP stub life, SB's , etc., supplemented by consideration of the customer data and individual Control & Accessory thresholds. Modules may remain time-continued unless a known condition requires action, e.g., LLP with inadequate stub life, High trending vibration, Module inspection finding outside EM limit, etc. Accessories may remain time-continued unless a known condition requires action, e.g., FADEC fault, AD application, visual/check inspection, etc.

On Condition Assessment

The on-condition assessment of the engine provides the OEM Propulsion Engineer with a status of the engine health prior to shop input. This allows the workscope definition to address hardware condition predicted by measured data. It also considers the hardware standard of the engine:

- Removal Reason: Why was engine removed and has engine reached expected TOW
- Trending Monitoring: Performance (EGT, etc.), Vibration, Oil Consumption
- Engine Events: Has the engine seen an in service "Event" or Operational Exceedance
- Borescope Results: Confirm modules are serviceable per AMM limits
- Chip Detector/Screens: Review Metallic particle findings
- LLP Life: Stub life on the Chapter 5 Life Limited parts, do they need replacing
- EDS: Consider new engine Service Bulletin embodiment
- Service Bulletins: Apply CUST# agreed SBs to the engine
- ADs: ADs applicable to engine.

Shop visit definitions

Shop visit

This definition is to be used for classifying removals

Shop Visit – an engine removal is classified as a shop visit whenever the subsequent engine maintenance performed prior to reinstallation entails:

- (1) a 360 degree separation of major mating engine flanges
- (2) the removal of a disk, hub, or spool.



Exceptions:

- Separation of a fan module from a propulsor (for transportation and handling purposes– i.e., GE90) is not a shop visit
- Replacement of an engine module on-wing is not a shop visit
- Replacement of a gear box is not a shop visit
- Accomplishment of a top and/or bottom case is not a shop visit
- Service Bulletin defines work performed as 'not a shop visit'

Engine Performance Restoration Shop Visit

The services performed during a shop visit in which , at a minimum, any one of the following modules is exposed disassembled and subsequently refurbished : high-pressure compressor, high-pressure turbine, or combustion chamber or stage one Low pressure turbine .

As required, based on engine condition and SV workscope objective, the PRSV can be any combination of one or more of these modules.

Typical Engine Performance Restoration Shop Visit

The services performed during a shop visit to insure optimum EGT levels where to do as a minimum the following modules are exposed, disassembled and subsequently refurbished :

- (i) High-pressure turbine and combustor chamber.
- (ii) Stage one low pressure turbine nozzle and the high-pressure compressor are workscooped on-condition, which may result in either a partial module workscope for cause , performance or overhaul as required, in order to achieve the next installation time on-wing or shop visit minimum build objectives.
- (iii) Fan, Low Pressure Compressor and Low Pressure Turbine module evaluation and cleaning.
Refurbishment of flowpaths and seals only if required based on condition of hardware if exposed.

Additional recommendation

Test or refurbishment of the EGT Indication System.

inspection and Test in accordance with the Component Maintenance Manual and refurbishment if required) of the Fuel Distribution and Airflow Control System components,

Different combination will be proposed in the LEAP MG workscope section.

LLP Shop visit

Scheduled shop visit dedicated to replace LLP parts.



Quick Turn Maintenance or Quick Turn Shop visit

A Quick Turn shop maintenance or Quick Turn shop visit are dedicated to do a partial off wing engine maintenance resulting to Quick Turn Around Time. Quick turn engine shop visits may omit category 3 service bulletins. Performance may or may not be affected by a Quick Turn SV depending on the modules affected or extent of workscope performed.

Continued Time Engine Maintenance (CTEM) and Continued Time Engine Shop visit

Engine Continued Time maintenance are dedicated to maintain the Engine in operation up to the LLP or expected typical Performance Shop visit. This engine maintenance at shop level may not increase the scheduled time to the next shop visit.

An engine continued time maintenance is a limited shop maintenance activity that results in the engine being continued time.

Example :

A Quick Turn maintenance to apply corrective actions.

An engine that has a shop maintenance for designated Service Bulletin (SB) activity.

An engine that is reworked or retrofitted for specific items, such as HPT blade replacement.

Depending the level of engine disassembly, this Continued Time maintenance activity can be declared as Shop visit or not. Continued time engine shop visits may omit category 3 service bulletins.

Level of work

LEAP Engine workscoping is defined to allow advantage to be taken of the flexibility that the "on-condition" maintenance philosophy provides. Each Engine is workscoped based on individual module requirements, thus allowing precise matching of shop workscope with Engine condition.

An Engine Workscope is an addition of workscopes instructions to define the work to be done at engine level, at major module level and at minor modules level.

Engine level and major module levels of work are mainly dedicated to give disassembly instructions to gain access to minor module. Minor modules instructions complete the workscope down to piece part level.

Four standard workscope levels of work (0, 1, 2, and 3) may be applied to each module to ease workscope understanding:

Level 00 Check Inspection / Level 01 Minimum workscope / Level 02 Performance or for Cause / Level 03 Full overhaul workscope.

Workscopes Levels references are

MMXX Level YY



MM: Major Module

SM: Shop Module

XX: Shop Module or Major Module Number

Level YY: Workscope Level of the selected module

Level 00: Check Inspection

Level 01: Minimum workscope

Level 02: Performance or for cause Workscope

Level 03: Full overhaul workscope

Workscopes Instructions at Engine level:

MM00 Level 00 Check Inspection

Level applicable for Continued Time Engine Maintenance Workscope on engine assembly

Refer to ESM 72-00-00 Check Inspection for overall condition

MM00 Level 01 Minimum Workscope with no Major Module Removal

The Engine Minimum Workscope constitutes the minimum maintenance requirements recommended to be completed at each shop visit on fully assembled Engine for scheduled Shop visit

This level highlights preventive maintenance and inspections required at this level of exposition and can identify where additional levels of work are required.

MM00 Level 02 Engine Workscope with one Major Module Removal

This level gives mainly instructions to partially disassemble engine with a major module removed.

This level introduces ESM Special procedure

MM00 Level 03 Engine Workscope with all 3 major modules split

This level gives mainly instructions to disassemble engine in 3 major modules.

Instructions at Major Module Level / Fan MM01 / Core MM02 / LPT MM03:

Major Modules are supposed to be removed for this level of workscope

MM01 (02/03) Level 00 Check Inspection

Level applicable for Continued Time Workscope on major module removed

Refer to ESM 72-00-00 Check Inspection for overall condition

MM01 (02/03) Level 01 Minimum Workscope

This level constitutes the minimum shop maintenance requirements to be completed at shop visit on each assembled Major Module assy removed from Engine.

This level highlights preventive maintenance and inspections and will identify if higher levels of work are required.

MM01 (02/03) Level 02 Partial Workscope with minor module removal

This level gives mainly instructions to partially disassemble major module into minor module. This level introduces ESM Special Procedure.

**MM01 (02/03) Level 03: Fully split into minor module**

This level gives main instructions to fully disassemble major module into minor modules.

Instructions at Minor Module Level: SM (ATA) Modules removed**Shop Module Level 00 Check Inspection**

- Applies to Continued Time engine maintenance dedicated to maintain the Engine in operation
- The module is removed and not disassembled then 72-00-00 check inspection criteria should be used.

Shop Module Level 01 Minimum Workscope

- The module is removed then 72-00-XX inspection criteria should be used.
- The assembled module workscope recommendations may apply at this level or minor disassembly for S/B application.
- Applies to all engine shop visits with modules having no apparent hardware failure or performance problems,
- Use diagnostics and intelligent engine trend monitoring and event reports to assist in workscope planning.
- This level gives recommended Service Bulletins to be applied for these workscope.

NOTE: This level highlights preventive maintenance and inspections required at this level of exposition and may require further module disassembly if an unserviceable condition is identified.

Shop Module Level 02 Performance or “for cause” Workscope

- Minimum workscope recommendations may apply at this level
- The “performance” workscope intended to restore EGT/SFC margin. It accomplishes a restoration of condition for the flowpath of the minor module
- The “for cause” Workscope is a curative or/and Preventative maintenance to enhance the reliability of the module.
- This level gives recommended Service Bulletins to be applied for these workscope.

NOTE: Minor modules will be considered partially dis-assembled. Example : rotors could be kept in the stacked configuration unless LLP or cause dictates unstacking.

Shop Module Level 03 Full overhaul workscope

- Assembled engine workscope and minimum Workscope recommendations may apply at this level.
- This workscope accomplishes a complete minor module overhaul. Entails thorough piece-part inspection and refurbishment per applicable Engine Shop Manual Chapters (including all life-limited part NDT Mandatory Inspections).
- Refurbishment of Items having the greatest potential for regaining EGT Margin is included in that Full Overhaul workscope
- Incorporate recommended Service Bulletins for this level



Shop Module Level 03 Performance / Full overhaul workscope

Some Shop module can have only a combined Level 03 Performance / Full overhaul when piece part level is necessary for performance restoration

Check inspection for Continued Time Engine Maintenance Workscopes

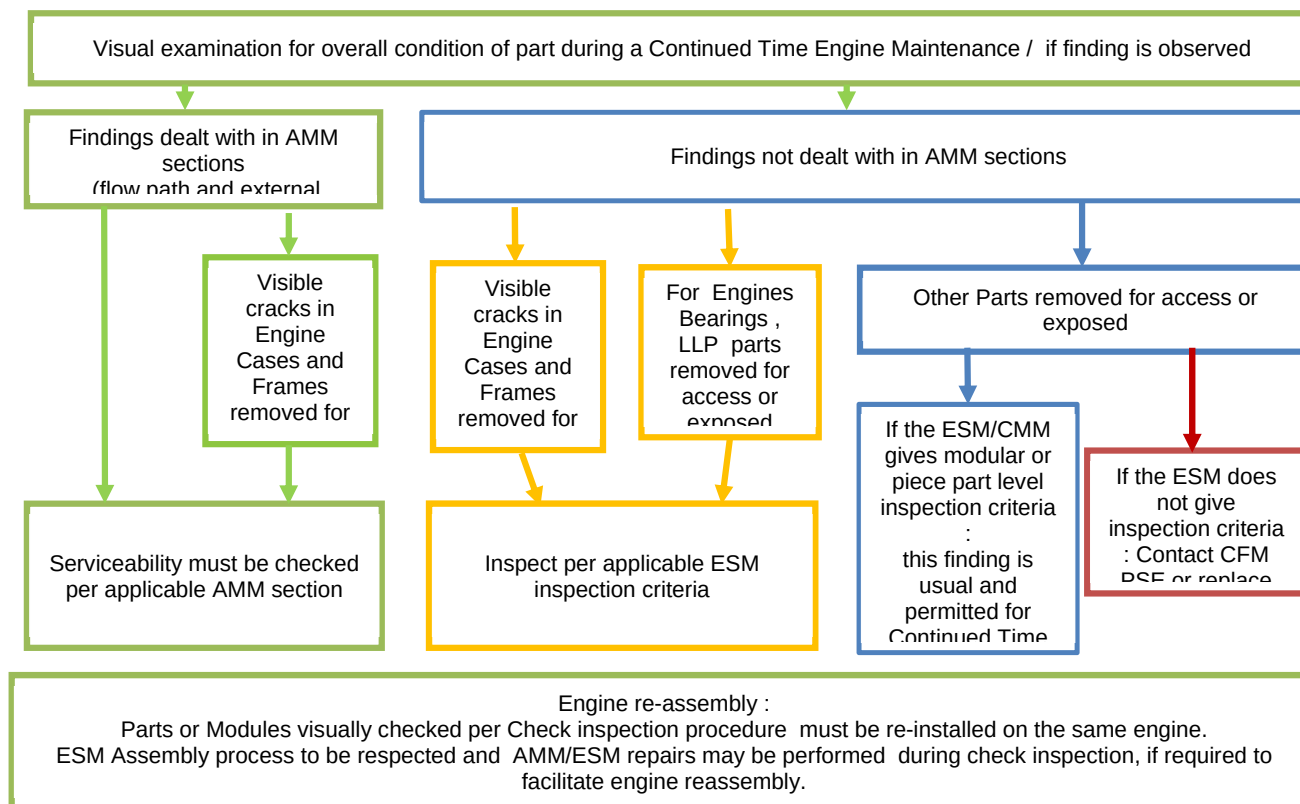
CFM has developed an ESM procedure defined at engine level 72-00-00 – Check Inspection to manage the concept of Continued Time Engine Maintenance Shop visit. This procedure allows to use AMM criteria at shop level to maintain engine in operation up to the next scheduled Shop visit and describes condition to do so.

This procedure gives instructions to do a check inspection for overall condition to detect structural failure or excessive deterioration and to determine the need for corrective maintenance of all visible engine parts:

(1) Do a visual examination for overall condition of any sub-assembly or part or control and accessory equipment removed for access or any area that becomes exposed by the removal of a sub-assembly or part or control and accessory equipment.

(2) During check inspection, a separate findings treatment is required

- Findings covered by AMM section, serviceability to be verified per appropriate AMM section
- Findings not covered by AMM section are considered as usual and acceptable for continued time service per ESM 72-00-00, CHECK INSPECTION if the ESM provides modular or piece part level inspection criteria.
- All other findings as well as findings on LLP's, bearings and visible cracks in cases/frames are considered as not acceptable for continued time service and need to be inspected per applicable AMM or ESM inspection criteria.





Cleaning

All removed parts must be sufficiently cleaned to permit adequate inspection. For Check Inspection, this may be wiping of the part to remove dirt and surface deposit..

Engine condition assessment before Continued Time Engine Maintenance Workscope definition

- A. Do the recommended maintenance identified in the engine manual (EM) Chapter 05 sections:
- B. Do an inspection of the engine condition before the induction to ensure workscope effectiveness:
 - (1) Make sure that the engine performance trends are satisfactory and that the engine will have adequate performance margin to return to a service condition.
 - (2) Make sure that the engine has Life Limited Parts (LLP) stub life consistent with post repair goals. Refer to TASK 05-xx-xx-200-801 (05-11-00, LIFE LIMITS 001).
 - (3) If possible, make sure that all engine related fault messages within the central maintenance computer (CMC) fault history or in the electronic engine control (EEC) and engine monitoring unit (EMU) non-volatile memory (NVM) have been reviewed and appropriate maintenance action identified for the engine and parts.

NOTE: TASK 05-xx-xx-200-801 (05-11-00, LIFE LIMITS 001) defines rotating and static structure parts that have life limitation in relation to the time in service.

QEC (Quick Engine Change) / BFE (Buyer Furnished Equipment):

LEAP MAINTENANCE GUIDE does not cover QEC and BFE for workscope preparation and recommendation. To be detailed by Customer Team during Workscope preparation.



Technical Information Statement

Any recommendations, limits, thresholds, etc..... referenced in this, or any other approved CFM documentation, were developed assuming that the engine, module or component is of original CFM design and has been maintained in accordance with CFM approved documentation and recommendations. The guidance in this document does not address parts or repairs supplied by third-party sources which have not been approved by CFM.

Critical Parts and Critical Influencing Parts

ATA Chapter 05-00-00 of the Engine Shop Manual lists the rotating parts and static structural parts identified by the Federal Aviation Administration (FAA) and the European Aviation Safety Agency (EASA) as Critical Parts. The Airworthiness Limitations contained in Sections 05-11 and 05-12 have been substantiated to the Certifying Authorities based on engineering analysis that assumes this product will be operated and maintained using the configurations, procedures, and inspections provided in the Instructions for Continued Airworthiness supplied with this product by the Type Certificate holder. For Critical Parts, and parts that influence Critical Parts, any repair, alteration or other maintenance not technically substantiated by the Type Certificate holder, or its licensees, or any replacement of such parts not supplied by the Type Certificate holder, or its licensees, can introduce differences that may materially affect these limits. CFM recommends the use of only CFM-approved replacement parts and repairs.

Included in ATA Chapter 05-00-00 is a list of Critical Influencing Parts (CIPs) - parts that experience has shown can directly or indirectly influence the boundary conditions of the lifing system used to determine the Critical Part Airworthiness Limitations. To ensure that Airworthiness Limitations published in Sections 05-11 and 05-12 remain valid, Critical Influencing Parts should be maintained in a manner consistent with the inspection and repair procedures contained in Chapter 72 of the engine manufacturer's shop manual.

CFM recommends recording each Critical Part's operating configuration with regards to its CIPs whenever assembled from piece-part level during shop-level maintenance with the following information as listed in Table 2 below in Figure 1. CFM further recommends using the format as shown in Figure 1 and to keep this sheet with the Critical Part's records until life expiration to document configuration of all CIPs.

**Critical Influencing Parts Installed with this LLP****Table 1: LLP**

Critical Part Nomenclature	Critical Part Part Number	Critical Part Serial Number	Critical Part TSN	Critical Part CSN	ESN Installed	Date Installed

Table 2: CIPs

Critical Influencing Part Nomenclature	IPC/ATA Ref	QTY Per Engine	QTY CFM- Approved*	QTY Not CFM- Approved	CIP Not Exposed

NOTES:

1. Produce one sheet for each LLP installed in the engine during shop visit
 2. Enter one row for each Critical Influencing Part for the appropriate Critical Part as listed in Chapter 05-00-00
- * New part or part repaired per CFM manuals and recommendations



Section 2 : LEAP 1B Engine management



Introduction

LEAP Operational experience has shown that extending the time/cycles between shop visits may optimize the cost of engine operation, along with balancing the cost of LLP replacement. Controlling the type and extent of the workscope performed at shop visit is where the operator is able to influence the time an engine remains on-wing before shop visit is required. The type and extent of workscope also influences the risk of exposing modules that are intended to be "continue time" at shop visit intended to update LLP in the low pressure systems only.

Preparation of a workscope for an incoming engine is critical to assuring a cost effective maintenance program. The first step of this process is defining the current condition of the engine. Without a proper assessment of the engine condition at removal, the workscope might not adequately address adverse engine conditions that may exist that could result in a premature return to the shop. The second step is establishing outbound goals for the engine that meet or exceed the desired on-wing life.

Factors which help define the present condition of an engine include the removal cause (high vibration, bearing failure, overtemperature, performance deterioration, etc.), time and cycles accumulated since last engine shop visit, performance trend analysis, observed hardware conditions, on-wing operational and maintenance history, cycle-limited parts, and inbound test run results. These should all be considered when establishing the level of workscope to be performed.

Outbound goals should be established to enhance on-wing longevity, and must take into account several time/cycle factors. These include the determination of a minimum stub life for cycle-limited parts, minimum performance margin based on a calculated on-wing life goal, and reliability/durability improvements, which will ease the on-wing inspection burden to the operator, while enhancing the reliability of the engine.

We have a breakdown of the shop visit's engine time since new by PSE Cause for each of the shop visit causes. Purpose is to help operators understand typical time on-wing for different causes.

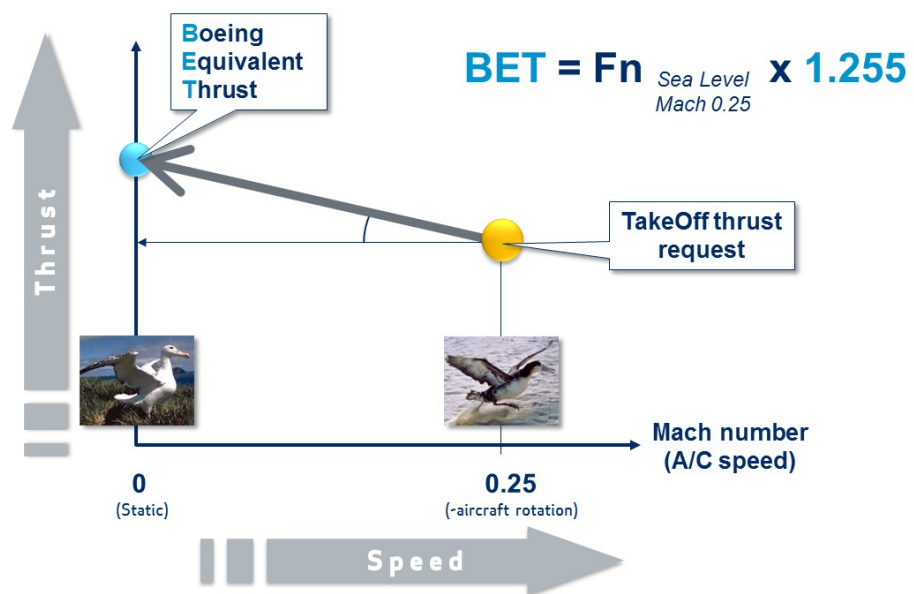


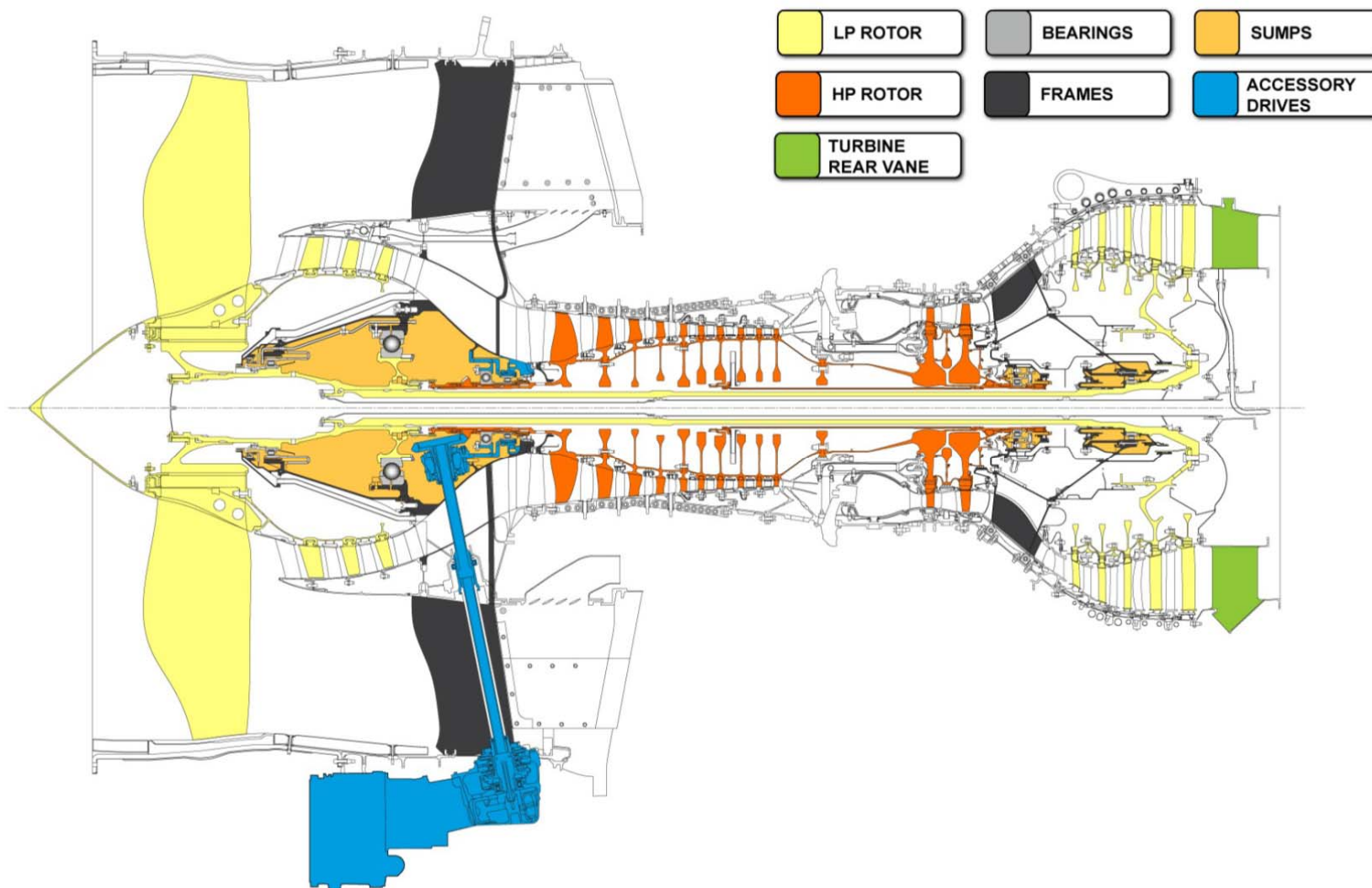
LEAP 1B Engine Family

Aircraft	Engine	BET	Nameplate	Description	T/O Thrust impact	EGTM impact
B 737 MAX-9	1B28B1	27,900	29,317	Hot day bump below 8400 ft	T/O Thrust Increase up to +6.6% vs B28 @ 7000ft Mach 0.25 ISA+30.6°C	
	1B28	27,900	29,317	ISA +15°C		
	1B27	26,400	28,037	Base -9 rating		
B 737 MAX-8	1B28B1	27,900	29,317	Hot day bump below 8400 ft Denver(5,324ft) Mexico(7,290 ft)	T/O Thrust Increase up to +6.6% vs B28 @ 7000ft Mach 0.25 ISA+30.6°C	
	1B28	27,900	29,317	ISA +15°C		
	1B27	26,400	28,037	ISA +15°C		
	1B25	25,000	26,786	Base -8 rating		
Boeing 737 MAX-7	1B27CB2	25,094	25,339	Thrust bump at High altitude Lhasa (11,711 ft)	T/O Thrust Increase up to +18% vs B27 from 10,000 ft to 15,000 ft	▼ -25°C vs. -1B27
	1B27C	25,094	25,339	Thrust reduction at Low altitude & Mach / ISA +15°C	T/O Thrust decrease down to-10% vs B27 @Sea Level Mach 0-0.2	
	1B25C	25,000	25,339	Thrust reduction at Low altitude & Mach / ISA +15°C	T/O Thrust decrease down to-6% vs B25@Sea Level Mach 0-0.2	
	1B23	24,000	25,907	ISA +15°C		
	1B21	23,000	25,014	Base -7 rating		



Boeing Equivalent thrust







LEAP 1B Engine description

1. Introduction

The engine assembly is a dual-rotor, variable stator, high by-pass ratio turbofan powerplant. The single stage fan and 3-stage booster are driven by a 5-stage **Low Pressure Turbine (LPT)**. The 10-stage **High Pressure Compressor (HPC)** is driven by a 2-stage **High Pressure Turbine (HPT)**. A second generation **Twin Annular Premixing Swirler (TAPS)** combustion chamber converts fuel and compressor discharge air into energy to give engine thrust and to drive the turbine. The accessory drive system gets energy from the **HPC** rotor to drive the engine accessories and the aircraft accessories that are installed on the engine. Reverse thrust for braking the aircraft after landing is supplied by an integrated system which acts on the fan discharge airflow.

2. Primary modules of the engine assembly

The primary modules of the engine assembly are as follows:

- A.** Compressor module consists of fan and booster module, No. 1 and No. 2 bearing support module, fan case module, fan frame module and **HPC** rotor/stator assembly.
- B.** Combustion module consists of combustion case assembly and combustion chamber assembly.
- C.** Turbine module consists of **HPT** nozzle, **HPT** rotor, **HPT** case, **LPT** nozzle module, **LPT** shaft module, **LPT** rotor and stator module and **Turbine Rear Vane (TRV)**.
- D.** Accessory drive module consists of **Inlet Gearbox (IGB)** & **Accessory Drive Train (ADT)**. **ADT** consists of **Transfer Gearbox (TGB)** assembly and **Accessory Gearbox (AGB)** assembly.

3. Detailed description of the modules

A. Compressor module

(1) The fan and booster module consists of a fan stage of 18 blades and a 3-stage axial booster, cantilever-mounted at the rear of the fan disk. The fan and booster module is located behind the air intake section. Through the fan disk, it is attached to and supported by the **Low Pressure Compressor (LPC)** shaft. The **LPC** shaft is driven by the **LPT**. The compressor module is attached and supported by the fan frame.

(2) The No. 1 and No. 2 bearing support module is located in the forward sump between the fan and booster module and the fan frame module. It is supported by the fan frame. It consists of a **LPC** shaft, two bearing supports, two bearings (front roller bearing and rear ball bearing), a load reduction device, a lubrication system, a front sealing system and the speed sensor wheel.

(3) The fan case module is located behind the air intake section and supported by the fan frame module. It consists of the fan case, an abradable assembly, a forward acoustic shroud and rear acoustic panels.

(4) The fan frame module has the engine mounting located at 12 o'clock on the rear flange of the fan hub. It consists of a fan frame shroud, a set of **Outlet Guide Vanes (OGV)**, an engine kit, a fan hub, a **Variable Bleed Valves (VBV)** door system and an oil distributor.



The main functions of the fan frame module:

- To direct and ensure a smooth aerodynamic airflow
- To support the engine
- To transmit power plant thrust to the aircraft.

(5) The **HPC** is located between the fan frame module and the **HPT** module. The shape of each compressor rotating blade incorporates the latest 3D-aero technology designed to efficiently move air to the next higher (rear) stage. The **HPC** is a 10-stage axial flow compressor.

(a) The **HPC** rotor with stages 1 to 5 blisk and stages 6 to 10 rotor spool. Stages 6 to 10 have axial dovetail slots. The rotor blades for each stage are attached with two locking lugs to prevent circumferential movement.

(b) The **HPC** front stator consists of the **Inlet Guide Vanes (IGV)** and the forward compressor stator case with four stages of **Variable Stator Vanes (VSV)**. Forward compressor stator case is divided into two matched halves with flanges at the 3 and 9 o'clock locations. Borescope ports are provided at each **HPC** stage for internal inspections.

(c) The **HPC** rear stator contains five stages of fixed compressor stator vanes. Borescope ports are provided at each **HPC** stage for internal inspections.

B Combustion module

(1) The combustion section is located between **HPC** and **HPT**. The combustion section consists of the combustion case assembly and the combustion chamber assembly. The combustion section mixes fuel and **HPC** air for efficient and uniform combustion. It is the structural interface, transmits the engine axial load and provides the gas flow path between the compressor and the **HPT**. The combustion case has four borescope ports for internal inspections of the combustion chamber assembly. The combustion case has a port to be used for reading pressure by the **Pressure Sub-system (PSS)** box of the control system.

(2) The combustion chamber assembly consists of 19 **TAPS** fuel nozzles, two igniters, combustion chamber dome, inner liner, outer liner, radial mixer and the inner and outer baffle. The **HPC** discharge air from the diffuser is used for cooling the combustion chamber.

C Turbine module

(1) The turbine section consists of the stage 1 **HPT** nozzle assembly, **HPT** rotor assembly, **HPT** case & stage 2 nozzle assembly, **Turbine Center Frame (TCF)** assembly, **LPT** stage 1 nozzle module, **LPT** shaft module, **LPT** rotor & stator module and **TRV** module. The main purpose of the turbine section is to provide rotational energy to the compressor section.

(a) The **HPT** module is located between the combustion section and the **TCF** assembly. The **HPT** is 2-stage turbine with 60 stage 1 blades and 68 stage 2 blades. The **HPT** extracts the energy from the hot gases exiting the combustion chamber, to drive the **HPC**, the **IGB** which drives the **TGB** and the **AGB**.

(b) The stage 1 **HPT** nozzles are physically located in the aft end of the combustion case. The stationary stage 1 **HPT** stator nozzles are needed to direct air to the **HPT** rotor blades. The stage 2 **HPT** nozzles contain holes visible from the flow path for cooling.

(c) The stage 1 **LPT** nozzle module is physically located between the aft end of the **TCF** and forward end of the **LPT** rotor and stator assembly.

(d) The **LPT** rotor and stator module is a 5-stage turbine. It is located between the **TCF** assembly and the **TRV** module. The **LPT** extracts the energy from the hot gases exiting the **HPT**, to drive the fan and booster rotor.



(e) The **TRV** is located at the end of **LPT** rotor & stator module. The functions of **TRV** to direct and ensure a smooth aerodynamic airflow.

(f) the **TCF** assembly support engine mount to the pylon rear attachment and cooling

D. Accessory drive module

(1) The engine and aircraft accessories are mounted on the **AGB** which is located on the lower portion of the fan casing and is driven by a shaft from the **TGB**. Power for the engine and the aircraft-mounted accessories is extracted from the **HPC** rotor shaft through an **IGB** through the **Radial Drive Shaft (RDS)** to the **TGB**.

(a) The **IGB** and the No. 3 bearing is located in the forward sump of the engine, secured on the fan frame assembly. The **TGB** and **AGB** are located on the fan frame at 8 o'clock location

4. Bearings and seals

A. Bearings

The engine rotors are supported by bearings installed in the sump cavities of the two frames. The A sump (forward) is in the fan frame and holds the bearings No. 1, No. 2 (fan shaft) and No. 3 (**HPC** front shaft forward part). The B sump holds the bearing No. 4 and the C sump holds the bearing No. 5.

B. Oil distribution

The bearings must be lubricated and oil is distributed to these components by nozzles. However, as the oil must be in the engine, seals of various types are installed to keep the oil and direct its recirculation.

C. Seals arrangement

The arrangement of oil and air seals, the provisions for oil supply, oil scavenge, seal pressurization and sump vent subsystems make a dry sump system. A sump is vented to ambient pressure through the front, rear and center vent tubes which are installed in the **LPT** shaft. The B sump and C sump are not vented.

D. Functions of the bearings

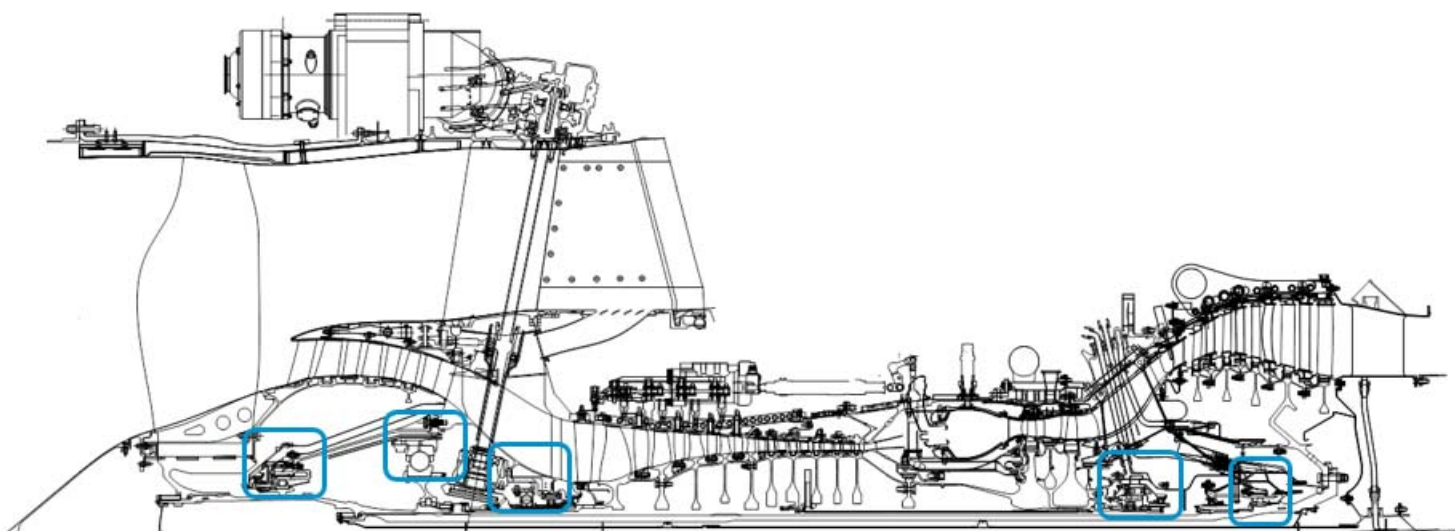
Bearings decrease the rolling friction, support the rotors axially and radially within the engine structure. The bearings hold the rotors relative to the stators. The bearing must control the forces of gravity weight, aerodynamic loads of pumping and turbine driving. The bearings also control the gyroscopic loads due to aircraft maneuvers.



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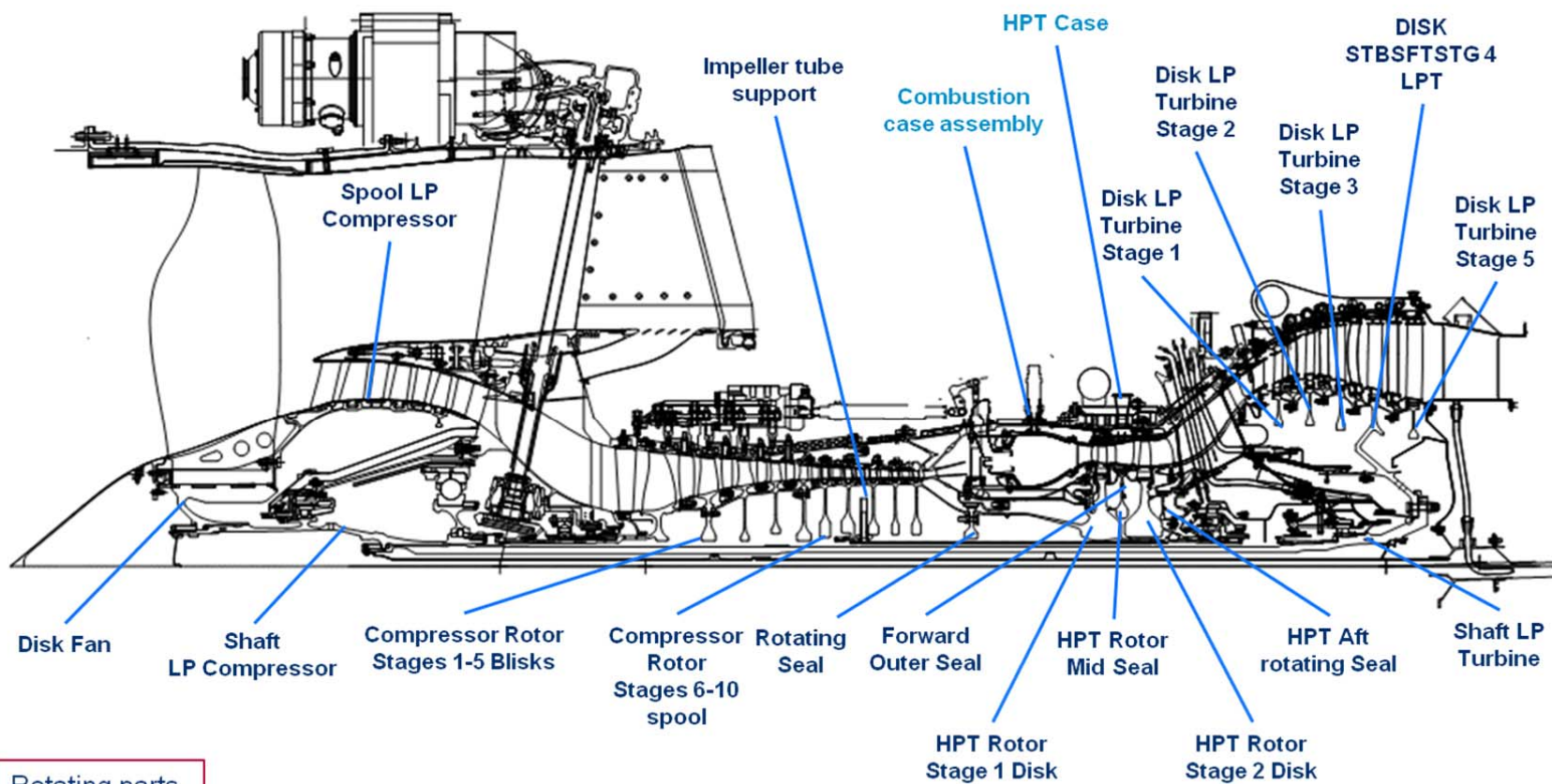
LEAP-1B main bearings



Improvements vs. CFM56	#1 bearing	#2 bearing	#3B bearing	#3R bearing	#4 bearing	#5 bearing
	• Roller bearing vs ball on CFM56 • Centered squeeze film damper	• Ball bearing vs roller on CFM56	• Ceramic ball • Nitrided metallic race • Under race lubrication • Added outer race drain holes	• Improved inner race material • Under race lubrication • Centered squeeze film damper	• Roller bearing • Non-intershaft bearing / Blind assembly eliminated	• Centered squeeze film damper
Benefits	Squeeze film damper reduces response to unbalance & engine vibration related noise Less variation in vibration with altitude due to roller at #1		Improved durability Improved heat rejection	Squeeze film damper reduces response to unbalance & engine vibration related noise Improved heat rejection	Reduced chance of assembly damage	Reduced response to unbalance & engine vibration related noise



LLP parts overview



Rotating parts
Static parts



Life Limited Part targets (Ultimate life)

Fan Disc	30,000 cycles
Fan forward shaft	30,000 cycles
Booster Spool	30,000 cycles
HPC Stage 1-5 Blisks	20,000 /17,500 Cycles 1B27 / 1B28
Stage 6-10 spool	20,000 /17,500 Cycles 1B27 / 1B28
CDP Seal	20,000 /17,500 Cycles 1B27 / 1B28
Forward outer Seal	20,000 /17,5 00 Cycles 1B27 / 1B28
HPT stage 1 disc	20,000 /17,500 Cycles 1B27 / 1B28
HPT stage 2 disc	20,000 /17,500 Cycles 1B27 / 1B28
LPT stage 1-5 discs	30,000 cycles
CDN , HPT casings	30,000 cycles

Extension to ultimate lives of 20,000/17,500* cycles (Core) and 30,000 cycles (Fan & LPT) within 5-6 years based on fleet utilization rates



Yearly LEAP 1B scorecards (June 2017)

LEAP 1B hours distribution by Shop visit Number (December XXX)

To Be Defined after EIS

LEAP 1B cycles distribution by Shop visit Number (December XXX)

To Be Defined after EIS

Shop Visit Top Causes - Last 12 months (December XXX)

Last 12 Months: Engine Flight Hours: ; Engine Flight Cycles:

To Be Defined after EIS

Delay and Cancellation Top Causes – Last 12 months (December)

Last 12 Months: Engine Flight Hours: ; Engine Flight Cycles:

To Be Defined after EIS



LEAP-1B EGT margin retention

To Be Defined after EIS

LEAP 1B Total SV rate (last 24 months)

To Be Defined after EIS

Leading indicator Program

To Be Defined after EIS

Shop Visit Key Workscope Items or TOP field Issue

To Be Defined after EIS

Potential Shop Visit Drivers



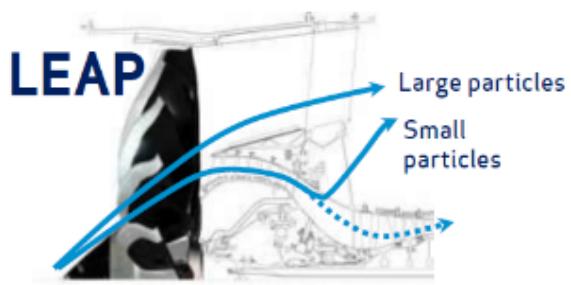
Maintenance recommendation in Severe Environment

Over the years CFM has developed several solutions to mitigate the in-service issues and increase engine TOW

- Specific regional borescope inspection program
- Engine preservation recommendations
- Water wash policy

LEAP Engines designed to improve core protection.

- Spinner directs debris into by-pass flow
- Wide chord fan centrifuges large particles to by-pass
- Inward opening VBV scoops small particles into by-pass



Most particles centrifuged away by wide chord fan blade
Spinner directs debris into bypass flow

Smaller particles exit through VBV doors

Rugged damage-tolerant HPC blisk technology

Reduced ice accretion, better ground shedding capability



Engine Protection

Use of Inlet/Outlet Covers overnight or during sand storms strongly recommended to all operators in Middle East/North Africa regions

Follow the AMM Engine Preservation Recommendations

Engine Inlet and Outlet must be Properly fixed regarding weather condition forecasted and Covers should be properly fixed from the front to the rear of the engine to avoid such sand accumulation

Engine Preservation

1. Preservation is required when an engine is removed for shop visit, staggering, or storage
 - Engine preservation is critical for prevention of corrosion on engine bearings and engine fuel/oil wetted components
 - Bearing spalling and failure has been attributed to corrosion on bearing races and balls or rollers.
2. In Middle East/North Africa protect the engine against sand ingestion with Fan inlet and exhaust covers
 - After volcanic ash, dust or sand ingestion refer to AMM

Engine Water Wash

Engine wash recommended :

1. To reduce dust/sand deposits on airfoils
2. Performance recovery (EGT Margin and Fuel Consumption)
3. To eliminate organic deposits after a bird strike

Use AMM recommendations to avoid detergent entering aircraft air system resulting smell in cabin

Use pure water for sand reduction and performance recovery

Use pure water and detergent to eliminate organic or mineral deposits

Water wash Equipment :

Criteria to be considered when choosing one system :

- Efficiency : Select the system that offer the water to go straight inside primary flow path + compressor running at min speed for dirty water elimination
- Environment : Check if residues going outside of the engine are collected or filtered
- Process : Make sure the system is appropriate and easy to use by your team (equipment availability, installation requirements and timing)

Engine Conversion Services Bulletin & documents

S/B Number	Title	Compliance Category	Description / Action Required
72-000X	LEAP 1B Models identification.	9	

LEAP 1B Conversion Chart





Shop visit readiness



Shop visit readiness

The following chapter intends to describe a Shop visit scenario expected for mature engines in operation in neutral environment

The following scenarios are based on TOW optimization and are examples for planning purposes only. Individual customer workscope policy on minimum LLP stub life and Operational condition should be considered

The engine figures shows the remaining LLP cycles and the color describe if module repair or partially repaired or if LLP replaced
For convenience and to ease demonstration, all the figures shows a single calculation for Cycles since new or since between SV cycle on wing values but for each individual engine cycles on wing will be inside the minimum and maximum expected

Theses expected values are given for the following average operational parameters :

Neutral environment (non-harsh), mature engine

Single FL ... 1.8h/FC

Utilization ... 3200 EFH per year

Derate 10%, 70°F ambient temp

Evolution of this parameters will modify the expected time on wing.



LEAP-1B21 : Neutral environment, Engine mature

Operational parameter

Neutral environment (non-harsh), mature engine
 Single FL ... 1.8h/FC
 Utilization ... 3200 EFH per year
 Derate 10%, 70°F ambient temp

Fan LLP: 30K cycles
 HPC LLP: 20K cycles
 HPT LLP: 20K cycles
 LPT LLP: 30K cycles





LEAP-1B23 : Neutral environment, Engine mature

Operational parameter

Neutral environment (non-harsh), mature engine
 Single FL ... 1.8h/FC
 Utilization ... 3200 EFH per year
 Derate 10%, 70°F ambient temp

Fan LLP: 30K cycles
 HPC LLP: 20K cycles
 HPT LLP: 20K cycles
 LPT LLP: 30K cycles

Module
Repaired
with LLPs

Module
Repaired
No LLPs

Module
Partial
Repair





LEAP-1B25 : Neutral environment, Engine mature

Operational parameter

Neutral environment (non-harsh), mature engine

Single FL ... 1.8h/FC

Utilization ... 3200 EFH per year

Derate 10%, 70°F ambient temp

Fan LLP: 30K cycles

HPC LLP: 20K cycles

HPT LLP: 20K cycles

LPT LLP: 30K cycles





LEAP-1B28 : Neutral environment, Engine mature

Operational parameter

Neutral environment (non-harsh), mature engine

Single FL ... 1.8h/FC

Utilization ... 3200 EFH per year

Derate 10%, 70°F ambient temp

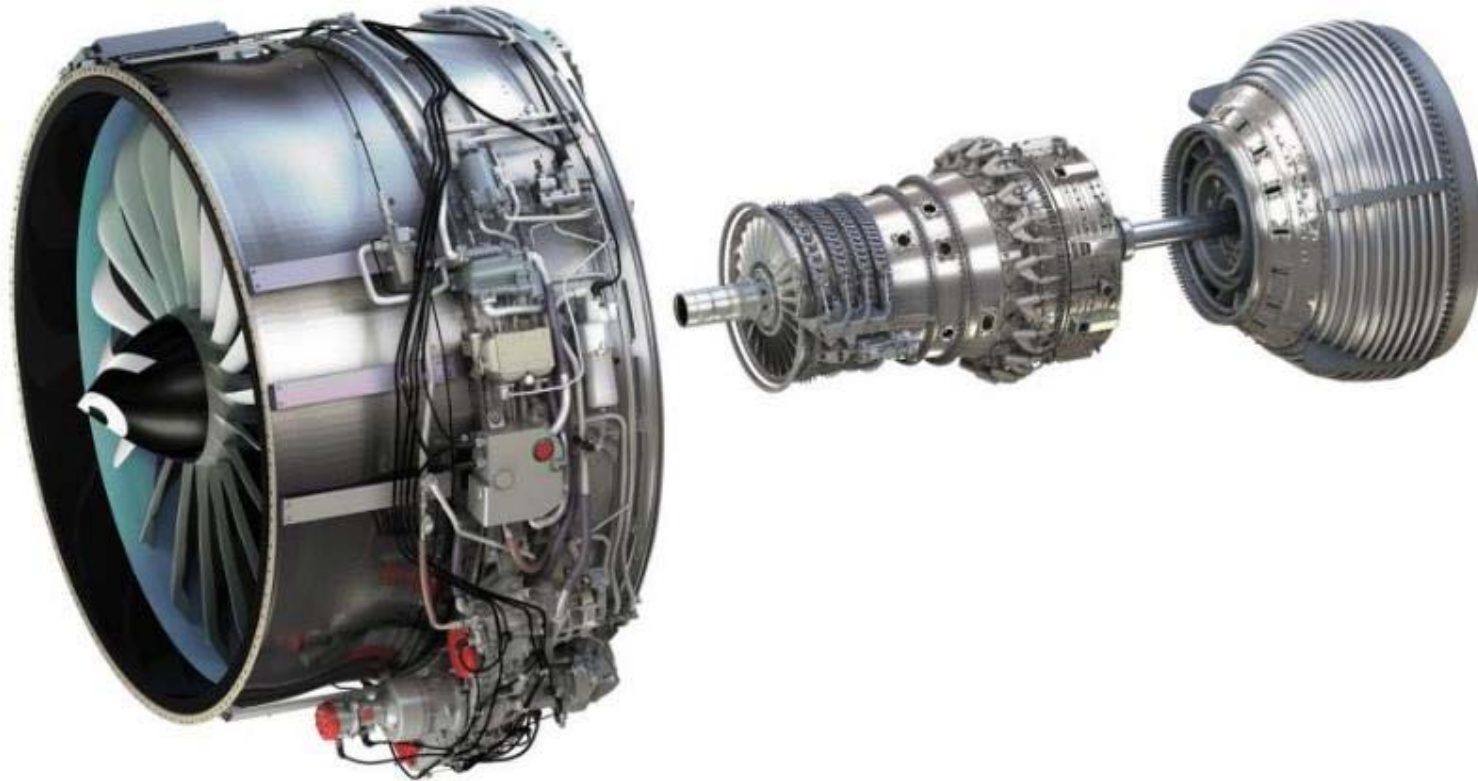
Fan LLP: 30K cycles

HPC LLP: 17,5K cycles

HPT LLP: 17,5K cycles

LPT LLP: 30K cycles







Section 3 : Engine Worksopes Digest





Introduction : Workscope Forecasting / Planning Factors

Some of the factors that can enter into workscope decisions are listed below. This is not an inclusive list and some of these situations may or may not enter into the workscope decisions. Each situation is unique.

LLP Remaining – Determines which modules need to be exposed for LLP replacement

LLP Stub Life Criteria – Be sure to “match” LLP lives across the engine or major module to minimize exposure at the next shop visit

Engine Configuration – Review S/B Status versus Maintenance Specification

Airworthiness Directives – Comply with at Shop Visit

Lease Return Requirements – EGT and LLP Requirements are significant driver to workscope

Ownership Horizon – Does it make sense to “gold plate” engine workscope if engine is to be returned to the lessor within a short time period?

Engine Distribution (TSV/CSV) – May need to build on-wing entitlement into engines to stagger fleet shop visit schedule in the future

Shop Loading/Manpower – How quickly do we need the engine back? May drive heavy vs. light workscope

Engine Reliability – What S/B's are to be incorporated at shop visit?

Environmental Factors (erosion, pollution, ambient LEAP temperature)

Performance Deterioration- Any upgrades available to reduce deterioration?

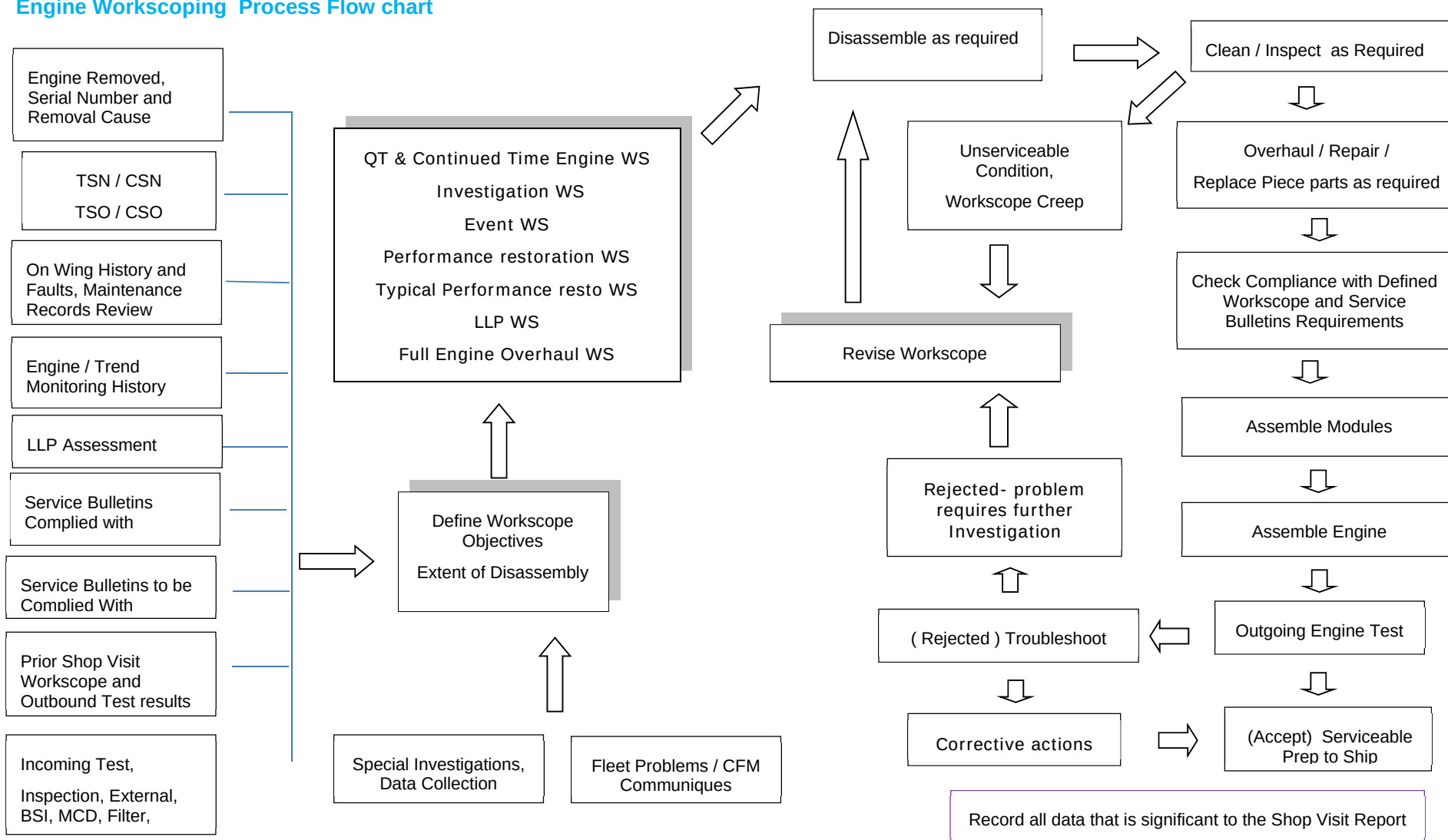
Derate Usage – This can impact deterioration rates

Engine Thrust – Any opportunity to re-rate engine – Impact on LLP life?

Water Wash Program – This can impact the engine TOW significantly. Operators should follow CFM recommendation for water wash interval.”



Engine Workscoping Process Flow chart



**LEAP MG : LEAP-1B example workscopes**

LEAP-1B Example Workscopes
Engine Performance restoration : HPT module repair
Engine Performance restoration : Hot Section (CC + HPT) repair
Engine performance restoration : Core engine minus LPT Stg 1 nozzle
Engine Performance restoration : Complete Core engine
Engine Performance restoration : Core with all core LLP minus LPT stg 1 nozzle
Engine Performance restoration : Complete Core with all core LLP
Engine Performance restoration : Core minus STG 1 LPT nozzle + Booster
Engine Performance restoration : Complete Core + Booster
Engine Performance restoration : Complete Core + Low pressure modules LLP
Engine Performance restoration : Engine Full overhaul with full LLP
Low Pressure Modules Overhaul with LLP
Booster Module Repair
LPT module Repair
Accessory Gearbox Repair
Transfer Gearbox Repair



Engine Performance restoration : HPT module repair

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	Performance Limitation or for Cause		
		Workscope designation	Engine Performance restoration : HPT module repair		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Partial	Level 01: Minimum Workscope Level 02.X : SM56 + SM 54 + SM 53 +SM 52 + SM 51 Removal /
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE			
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE			
Mm06	SM 31	HPC ROTOR			
Mm07	SM 32	FWD COMPRESSOR STATOR			
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE			
Mm09	SM 42	COMBUSTION CHAMBER			
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
ACCESSORY DRIVES					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
C & A			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration: Hot Section (CC + HPT) repair

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	Performance limitation or for cause		
		Workscope designation	Engine Performance restoration : Hot Section (CC + HPT) repair		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Partial	Level 01: Minimum Workscope Level 02.X: SM56 + SM 54 + SM 53 +SM 52 + SM 51 + SM42 Removal / Installation
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE			
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE			
Mm06	SM 31	HPC ROTOR			
Mm07	SM 32	FWD COMPRESSOR STATOR			
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE			
Mm09	SM 42	COMBUSTION CHAMBER	Removed	As required	Level 01: Minimum Workscope + Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	As required	Level 01: Minimum Workscope + Level 02.x For cause TBD or Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
ACCESSORY DRIVES					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
C & A			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine performance restoration : Core engine minus LPT Stg 1 nozzle

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	Performance limitation		
		Workscope designation	Engine performance restoration : Core engine minus LPT Stg 1 nozzle		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Partial	Level 02 : Performance STG 6-10 HPC Blade Replacement
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	N/A	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Partial	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed		Level 01: Minimum Workscope
LPT Minor Modules					
Mm20	SM 55	N05 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			On condition : Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration : Complete Core engine

LEAP Maintenance Guide WS digest					
LEAP 1B	Shop Visit Cause		Performance limitation		
	Workscope designation		Engine Performance restoration : Complete Core engine		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Partial	Level 02 : Performance STG 6-10 HPC Blade Replacement
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	N/A	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration : Core with all core LLP minus LPT stg 1 nozzle

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	LLP core engine		
		Workscope designation	Engine Performance restoration : Core with all core LLP minus LPT stg 1 nozzle		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	NLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	Full	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed		Level 01: Minimum Workscope
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration :
Complete Core with all core LLP

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	LLP core engine		
		Workscope designation	Engine Performance restoration : Complete Core with all core LLP		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	NLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	Full	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration : Core
minus STG 1 LPT nozzle + Booster

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	Performance limitation		
		Workscope designation	Engine Performance restoration : Core minus STG 1 LPT nozzle + Booster		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed		Level 01: Minimum Workscope Level 1.0X Booster module removal
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01 : Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER	Removed	Partial	Level 02 : Performance
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Partial	Level 02 : Performance STG 6-10 HPC Blade Replacement
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	N/A	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Performance / Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed		Level 01: Minimum Workscope
LPT Minor Modules					
Mm20	SM55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
ACCESSORY DRIVES					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
C & A			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration : Complete Core + Booster

LEAP Maintenance Guide WS digest					
LEAP 1B	Shop Visit Cause		Performance limitation		
	Workscope designation		Engine Performance restoration : Complete Core + Booster		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed		Level 01: Minimum Workscope Level 1.0X Booster module removal
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER	Removed	Partial	Level 02 : Performance
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Partial	Level 02 : Performance STG 6-10 HPC Blade Replacement
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	N/A	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
C & A			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration :
Complete Core + Low pressure modules
LLP

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	LLP and Performance		
		Workscope designation	Engine Performance restoration : Complete Core + Low pressure modules LLP		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY			Level 01: Minimum Workscope Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed		Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM2	72-00-02	CORE MAJOR MODULE	Removed		Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed		Level 01: Minimum Workscope Level 03: Split into Minor Modules
FAN & BOOSTER Minor Modules					
		FAN BLADE		Full	Fan Blades Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER	Removed	Full	Level 03 : Full Overhaul LLP
Mm02	SM 22	No1 and No2 BEARING SUPPORT	Removed	Full	Level 03 : Full Overhaul LLP
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING	Removed	Full	Level 03 : Full Overhaul LLP
	SM 23	FAN CASE			Level 01: Minimum Workscope WS
Mm04	SM 24	FAN FRAME			Level 01: Minimum Workscope WS
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Partial	Level 02 : Performance STG 6-10 HPC Blade Replacement
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	N/A	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE	Removed	Full	Level 03 : Full Overhaul
Mm16	SM 57	LPT SHAFT	Removed	Full	Level 03 : Full Overhaul LLP
Mm15	SM 58	LPT ROTOR AND STATOR	Removed	Full	Level 03 : Full Overhaul LLP
Mm17	SM 59	TURBINE REAR VANE	Removed	Full	Level 03 : Full Overhaul
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Engine Performance restoration : Engine Full overhaul with full LLP

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	LLP and Performance		
		Workscope designation	Engine Performance restoration : Engine Full overhaul with full LLP		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY		Full	Level 01: Minimum Workscope Level 02.X AGB Equipped Removal Level 02.X TGB Equipped Removal Level 03: Split into Major Modules
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM2	72-00-02	CORE MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
MM3	72-00-03	LPT MAJOR MODULE	Removed	Full	Level 01: Minimum Workscope Level 03: Split into Minor Modules
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Fan Blades Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER	Removed	Full	Level 03 : Full Overhaul LLP
Mm02	SM 22	No1 and No2 BEARING SUPPORT	Removed	Full	Level 03 : Full Overhaul LLP
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING	Removed	Full	Level 03 : Full Overhaul LLP
	SM 23	FAN CASE	Removed	Full	Level 03 : Full Overhaul LLP
Mm04	SM 24	FAN FRAME	Removed	Full	Level 03 : Full Overhaul LLP
Core HP minor Modules					
Mm05	SM 30	HPC MODULE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm06	SM 31	HPC ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm07	SM 32	FWD COMPRESSOR STATOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm09	SM 42	COMBUSTION CHAMBER	Removed	Full	Level 03: Performance / Full overhaul WS
Mm10	SM 51	STAGE 1 HPT NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
Mm11	SM 52	HPT ROTOR	Removed	Full	Level 03: Performance / Full overhaul WS
Mm12	SM 53	STAGE 2 HPT NOZZLE	Removed	Full	Level 03: Full overhaul WS
Mm13	SM 54	TURBINE CENTER FRAME	Removed	Full	Level 03: Full overhaul WS
Mm14	SM 56	LPT STAGE 1 NOZZLE	Removed	Full	Level 03: Performance / Full overhaul WS
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE	Removed	Full	Level 03 : Full Overhaul
Mm16	SM 57	LPT SHAFT	Removed	Full	Level 03 : Full Overhaul LLP
Mm15	SM 58	LPT ROTOR AND STATOR	Removed	Full	Level 03 : Full Overhaul LLP
Mm17	SM 59	TURBINE REAR VANE	Removed	Full	Level 03 : Full Overhaul
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX	Removed	Full	Level 03 : Full Overhaul
Mm19	SM 63	ACCESSORY GEARBOX	Removed	Full	Level 03 : Full Overhaul
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Low Pressure Modules Overhaul with LLP

LEAP Maintenance Guide WS digest					
LEAP 1B	Shop Visit Cause		LLP		
	Workscope designation		Low Pressure Modules Overhaul with LLP		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY		Partial	Level 01: Minimum Workscope Level 02: LPT module removal
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed	Full	Level 03: Split into Minor Modules
MM2	72-00-02	CORE MAJOR MODULE			Level 01: Minimum Workscope
MM3	72-00-03	LPT MAJOR MODULE	Removed	Full	Level 03: Split into Minor Modules
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Fan Blades Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER	Removed	Full	Level 03 : Full Overhaul LLP
Mm02	SM 22	No1 and No2 BEARING SUPPORT	Removed	Full	Level 03 : Full Overhaul LLP
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING	Removed	Full	Level 03 : Full Overhaul LLP
	SM 23	FAN CASE	Removed		Level 01: Minimum Workscope WS
Mm04	SM 24	FAN FRAME	Removed		Level 01: Minimum Workscope WS
Core HP minor Modules					
Mm05	SM 30	HPC MODULE			
Mm06	SM 31	HPC ROTOR			
Mm07	SM 32	FWD COMPRESSOR STATOR			
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE			
Mm09	SM 42	COMBUSTION CHAMBER			
Mm10	SM 51	STAGE 1 HPT NOZZLE			
Mm11	SM 52	HPT ROTOR			
Mm12	SM 53	STAGE 2 HPT NOZZLE			
Mm13	SM 54	TURBINE CENTER FRAME			
Mm14	SM 56	LPT STAGE 1 NOZZLE			
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE	Removed	Full	Level 03 : Full Overhaul
Mm16	SM 57	LPT SHAFT	Removed	Full	Level 03 : Full Overhaul LLP
Mm15	SM 58	LPT ROTOR AND STATOR	Removed	Full	Level 03 : Full Overhaul LLP
Mm17	SM 59	TURBINE REAR VANE	Removed	Full	Level 03 : Full Overhaul
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



Booster Module Repair

LEAP Maintenance Guide WS digest					
LEAP 1B	Shop Visit Cause		Booster module reliability		
	Workscope designation		Booster Module Repair		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY		Partial	Level 01 minimum workscope
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE	Removed	Partial	Level 01 minimum workscope Level 02.X: FAN & BOOSTER module Removal
MM2	72-00-02	CORE MAJOR MODULE			
MM3	72-00-03	LPT MAJOR MODULE			
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Partial	Level 01 minimum workscope
Mm01	SM 21	FAN AND BOOSTER	Removed	Full	Level 03 : Full Overhaul workscope or level 2.X for cause TBD
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE			
Mm06	SM 31	HPC ROTOR			
Mm07	SM 32	FWD COMPRESSOR STATOR			
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE			
Mm09	SM 42	COMBUSTION CHAMBER			
Mm10	SM 51	STAGE 1 HPT NOZZLE			
Mm11	SM 52	HPT ROTOR			
Mm12	SM 53	STAGE 2 HPT NOZZLE			
Mm13	SM 54	TURBINE CENTER FRAME			
Mm14	SM 56	LPT STAGE 1 NOZZLE			
LPT Minor Modules					
Mm 20	SM 55	N°5 BEARING SUPPORT MODULE			
Mm16	SM 57	LPT SHAFT			
Mm15	SM 58	LPT ROTOR AND STATOR			
Mm17	SM 59	TURBINE REAR VANE			
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



LPT module Repair

LEAP Maintenance Guide WS digest					
LEAP 1B		Shop Visit Cause	LPT ROTOR STATOR FINDINGS		
		Workscope designation	LPT module Repair		
MM / Mm	ATA	Level of work	Removal	Disassy	Workscopes Levels designation
Engine & Major modules					
ENGINE	72-00-00	ENGINE ASSEMBLY		Partial	Level 01: Minimum Workscope Level 02: LPT module removal
MM1	72-00-01	FAN & BOOSTER MAJOR MODULE			Level 01: Minimum Workscope
MM2	72-00-02	CORE MAJOR MODULE			Level 01: Minimum Workscope
MM3	72-00-03	LPT MAJOR MODULE	Removed	Full	Level 03: Split into Minor Modules
FAN & BOOSTER Minor Modules					
		FAN BLADE	Removed	Full	Fan Blades Level 03: Full overhaul WS
Mm01	SM 21	FAN AND BOOSTER			
Mm02	SM 22	No1 and No2 BEARING SUPPORT			
Mm03	SM 61	INLET GEARBOX AND No 3 BEARING			
	SM 23	FAN CASE			
Mm04	SM 24	FAN FRAME			
Core HP minor Modules					
Mm05	SM 30	HPC MODULE			
Mm06	SM 31	HPC ROTOR			
Mm07	SM 32	FWD COMPRESSOR STATOR			
Mm08	SM 41	COMBUSTOR DIFFUSER NOZZLE			
Mm09	SM 42	COMBUSTION CHAMBER			
Mm10	SM 51	STAGE 1 HPT NOZZLE			
Mm11	SM 52	HPT ROTOR			
Mm12	SM 53	STAGE 2 HPT NOZZLE			
Mm13	SM 54	TURBINE CENTER FRAME			
Mm14	SM 56	LPT STAGE 1 NOZZLE			
LPT Minor Modules					
Mm20	SM 55	N°5 BEARING SUPPORT MODULE	Removed		Level 01: Minimum Workscope
Mm16	SM 57	LPT SHAFT	Removed		Level 01: Minimum Workscope
Mm15	SM 58	LPT ROTOR AND STATOR	Removed	Full	Level 03 : Full Overhaul
Mm17	SM 59	TURBINE REAR VANE	Removed		Level 01: Minimum Workscope
Accessory Drives					
Mm18	SM 62	TRANSFER GEARBOX			
Mm19	SM 63	ACCESSORY GEARBOX			
Controls & Accessories			Refer to LEAP Maintenance Guide recommendation		
Test			Test 003 performance		



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Section 4: Services Bulletin Guidelines



Service Bulletin

To assist the operator in the determination of the important Service Bulletins to consider at a shop visit, condensed lists which highlights the Service Bulletins to be incorporated on an accelerated basis have been developed and included for each modules in its corresponding chapter. It is not all inclusive; reference should be made to the complete Service Bulletin list for recommendations on each Bulletin.

Service Bulletin Compliance key definitions:

The following standard categories are used for all LEAP 1B series Service Bulletins:

- **Category 1 : Usual Statement : CFM recommends that you do this Service Bulletin before subsequent flight or before xx hours, yy cycles or a specific end date or specified interval.**

Explanation: Compliance is mandatory, generally as a result of EASA or FAA action, i.e. **Airworthiness Directive (AD)** , **Notice of Proposed Rule Making (NPRM)** or pending **AD** etc. will cause customer action.

This category does require action within a specified time frame and must be complied with on all engines presently in the shop.

- **Category 2 : Usual Statement: CFM recommends that you do this Service Bulletin as soon as possible without effect on revenue service but before xx hours, yy cycles but a specific end date or specified interval.**

Explanation: Used when CFM technical reasons make the compliance necessary and when an aircraft can stay for suitable time at a line station or maintenance base with the capability to do the procedure. The justification for hour, cycle and end date requirements will be based on technical considerations only (i.e. safety, risk analysis)

This category may cause non-routine operator action and must be complied with on all engines presently in the shop.

- **Category 3 : Usual Statement: CFM recommends that you do this Service Bulletin at next shop visit of engine or module.**

Explanation:

Following an engine removal, it is classified as a shop visit whenever the subsequent engine maintenance performed prior to reinstallation entails a 360° separation of major mating engine flanges or the removal of a disk, hub or spool.

Exceptions:

- Replacement of an engine module on-wing,
- Replacement of a gear box,
- Accomplishment of a top/bottom case removal procedure,
- Service Bulletin defines work performed as not a shop visit.

Quick turn or continued time engine shop visits may omit category 3 service bulletins.



Disassembly of engine or module may be required to incorporate bulletin. Incorporation of this SB will enhance engine reliability and reduce the potential for operation disruptions. May cause further disassembly than planned.

• **Category 4 : Usual Statement: CFM recommends that you do this Service Bulletin when the area is exposed.**

Explanation: Technical evaluation indicates that incorporation of this SB may enhance engine operational reliability or improve time on-wing. Disassembly of a module to piece part may be required to incorporate this bulletin.

• **Category 5 : Usual Statement: CFM recommends that you do this Service Bulletin as soon as the affected part is removed from the engine.**

Explanation: Do when the affected part is removed at the piece-part level.

Technical evaluation indicates that incorporation of this SB is expected to improve (reduce) maintenance costs.

• **Category 6 : Usual Statement: CFM recommends that you do this Service Bulletin when the affected part is routed for repair.**

Explanation: Do when the affected part is removed and a repair is planned.

Incorporation of this SB should improve maintenance costs in the future.

• **Category 7 : Usual Statement: CFM recommends that you do this Service Bulletin at customer convenience/option.**

Explanation:

Do at customer convenience after the old parts are all used.

Do at customer option if old parts will be supplied (date supported) for engines that have not been changed.

An alternate configuration/part is being introduced into production that may have a different appearance or may require unique assembly or maintenance practices.

• **Category 8 : Usual Statement: Spare parts information**

Explanation: Use for release/introduction of spare parts that are fully interchangeable and are not anticipated to improve reliability or maintenance costs.

• **Category 9 : Usual Statement: Information only**

Explanation: Issued to announce specific name brand information (approved oils, etc.) or for administrative purposes.

S/B Manufacturer recommendations

One of the manufacturer recommendations that follow may be used in Management Information as compliance category:

- No manufacturer recommendation,



- Recommended,
- Desirable,
- Optional.

Compliance paragraphs of Service Bulletins with CFM cost participation will have two sections:

- Technical Compliance:

The technical compliance section will include the compliance categories and a CFM compliance statement.

Use of end dates on technical compliance will be based only on technical assessments.

- Program Support (Optional):

CFM may include a program support section to give financial participation data. This data will identify program financial end dates and indicate program details that can be found in materials and labor paragraphs.

Service Bulletin impact Statements :

The goal of the impact statement is to facilitate customers' prioritization of critical Service Bulletins to mitigate future significant events.

Impact letters Impact statements

A This recommendation is to address a condition that may affect flight safety.

B This recommendation is to address a condition that may result in an increased rate of **In-Flight Shutdowns (IFSD)** , **Take-Off Aborts (TOA)** , **Air Turn Backs (ATB)** or **Diversion (DIV)** .

C This recommendation is to address a condition that may result in a non-event operational disruption, **Unscheduled Engine Removals (UER)** , out-station-removals or **Aircraft On Ground (AOG)** .

D This recommendation is to address a condition that may result in a **Delay and Cancellation (D&C)** .

E This recommendation is to improve the cost of ownership, reduce maintenance requirements or is a product improvement.

F Implement as deemed necessary per the Service Bulletin category.



On wing S/B list

On Wing Maintenance Program Checking
Cat 1 & 2 S/B to be done on wing by Airline / Customer

ATA	S/B #	S/B designation	Compl	Impact statements	S/B Description and PSE recommendation

Borescope Inspection (BSI)

At shop incoming, perform a borescopic inspection per AMM of modules expected to be unexposed during this SV.

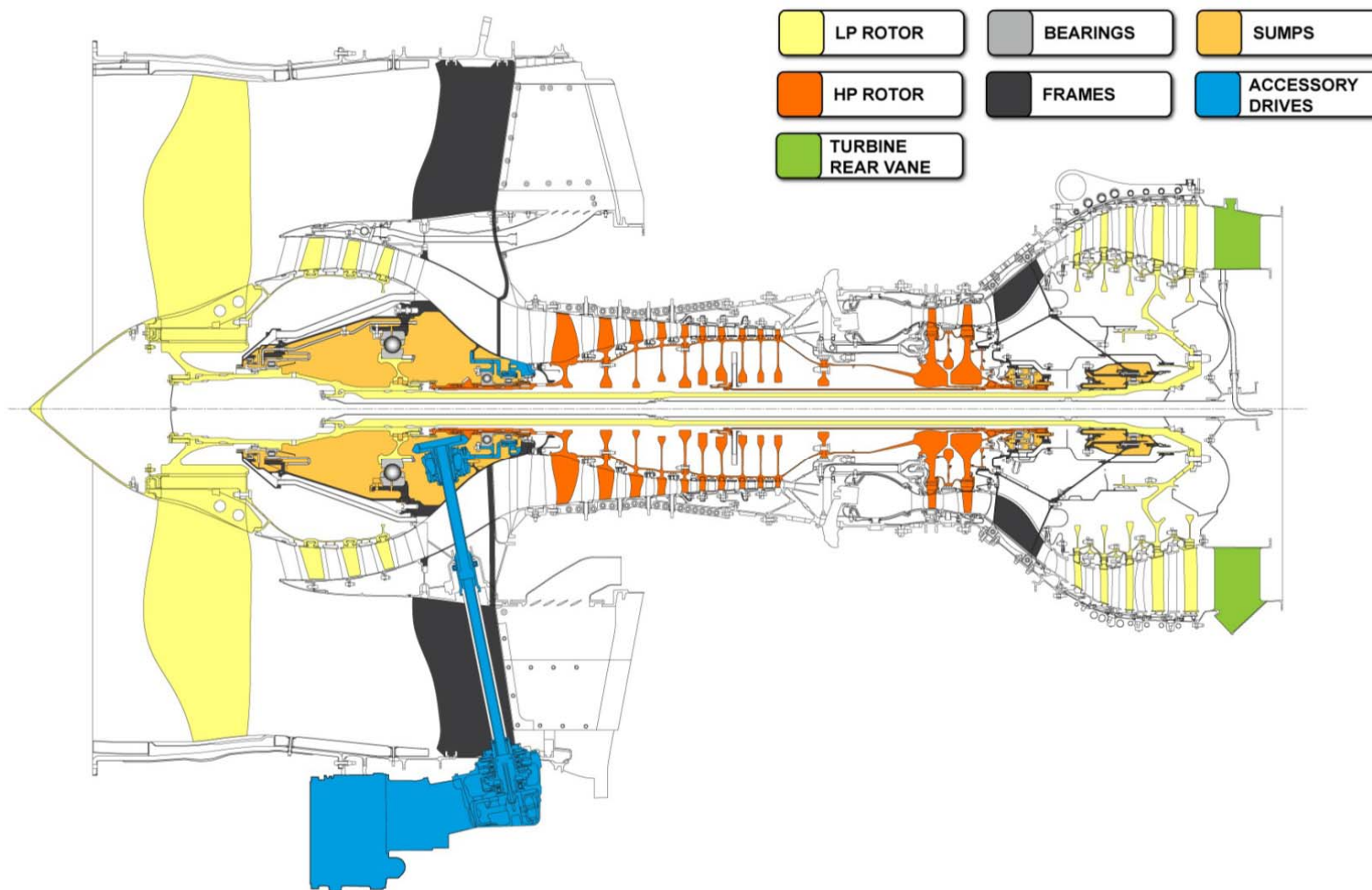
On modules concerned by incoming BSI, following BSI SB can be applied if required as per customer decision

ATA	S/B #	S/B designation	Compliance	Impact statements	S/B Description and PSE recommendation



MM00 – Engine assembly







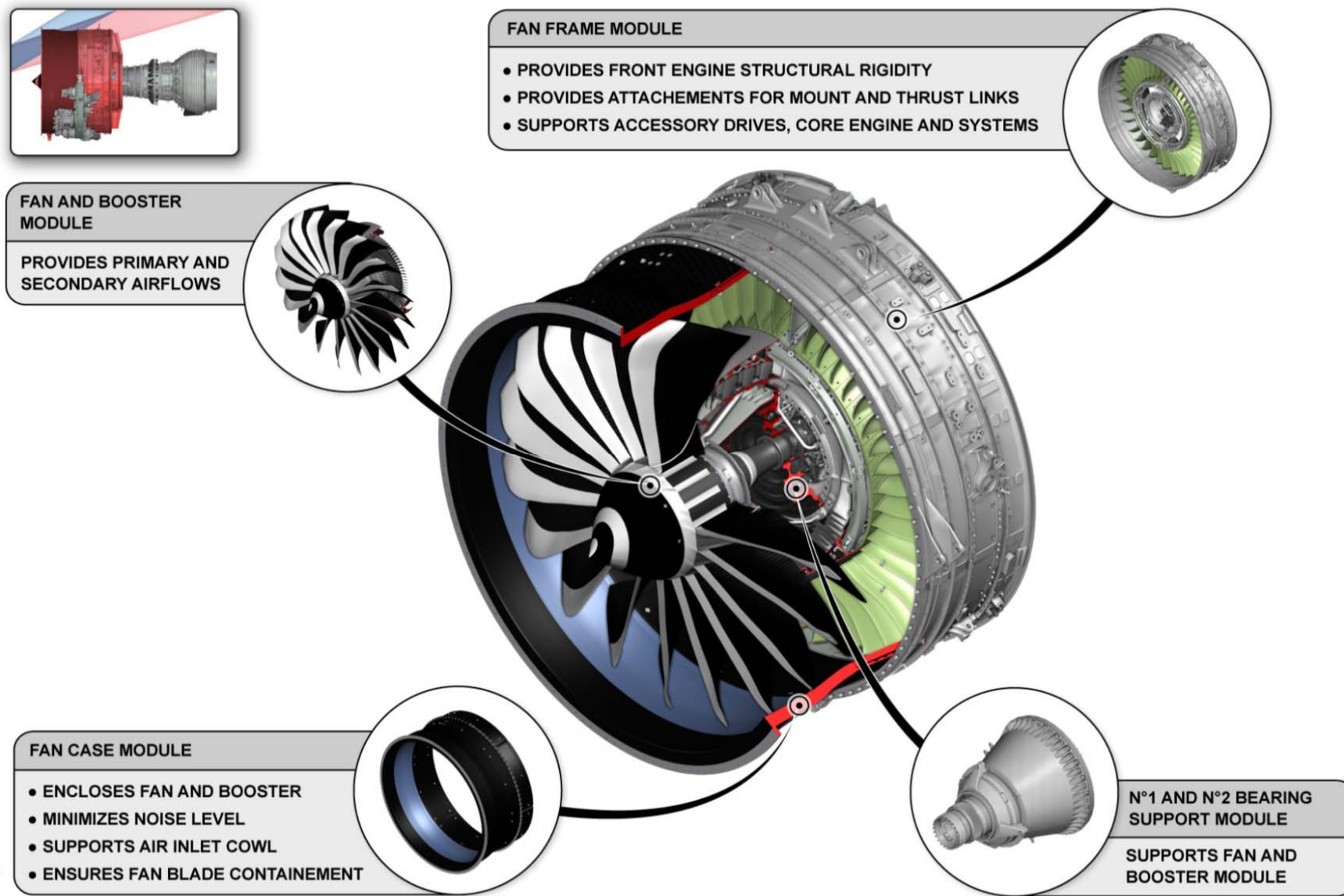
ENGINE ASSEMBLY – MM00 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation
72-002X	Inspection of ...	Cat 1	A	Ranking per Cat + ranking per Impact letters + ranking per S/B number or PSE recommendation
72-002X	Highest category first on the list	Cat 1	B	
72-00XX		Cat 2	B	
72-00XX		Cat 2	...C,D,E	
72-00XX		Cat 3	B	
72-00XX		Cat 3	...C,D,E,F	
72-00XX		Cat 4	B	
72-00XX		Cat 4	C,D,E,F	
72-00XX		Cat 5	B	
72-00XX			C,D,E,F	
72-00XX		Cat 6	B C,D,E,F	
72-00XX		Cat 7	C	
72-00XX			D	
72-00XX			E	
72-00XX			F	



MM01 – Fan Major Module

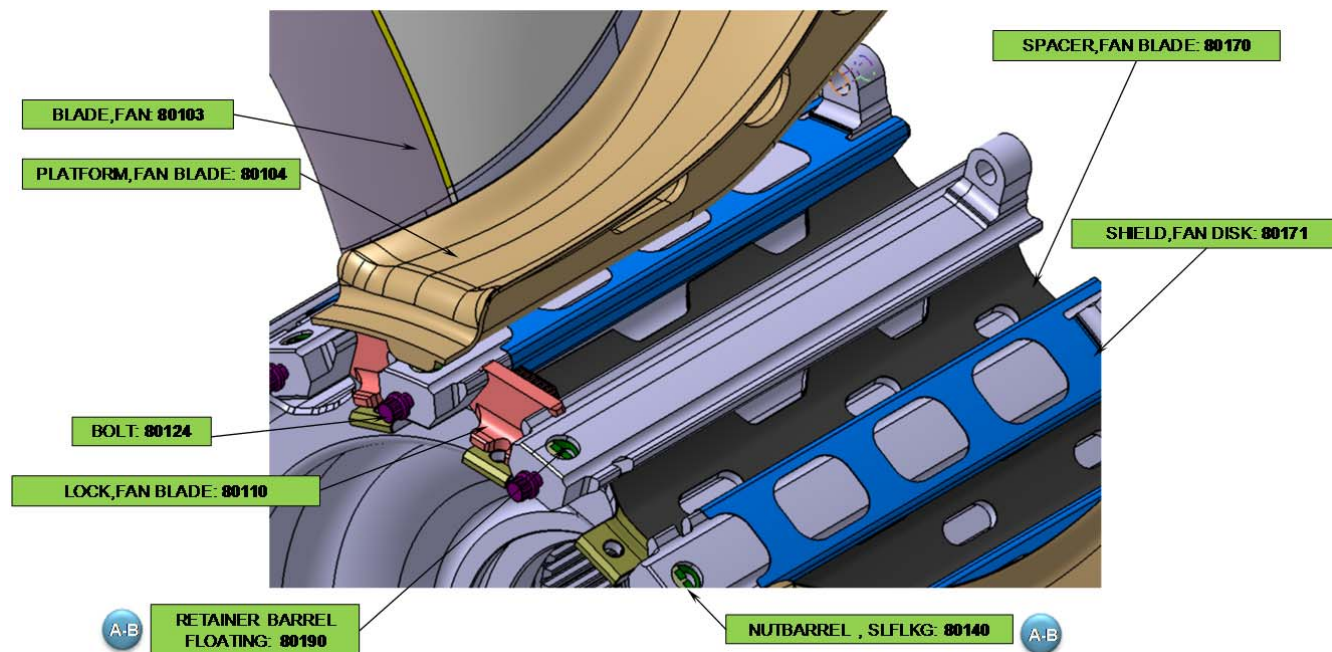






FAN MAJOR MODULE – MM01 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Fan Blades and associated hardware



FAN BLADES – SM21 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM21 – Fan and Booster





FAN AND BOOSTER – SM21 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM22 – No.1 and No.2 Bearing Support





N°1 & N°2 BEARING SUPPORT – SM22 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM23 – FAN CASE





FAN CASE – SM23 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM24 – FAN FRAME





FAN FRAME – SM24 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM61 - INLET GEARBOX AND No 3 BEARING





INLET GEARBOX & N°3 BEARING – SM61 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



MM2 72-00-02 – Core Engine Major Module

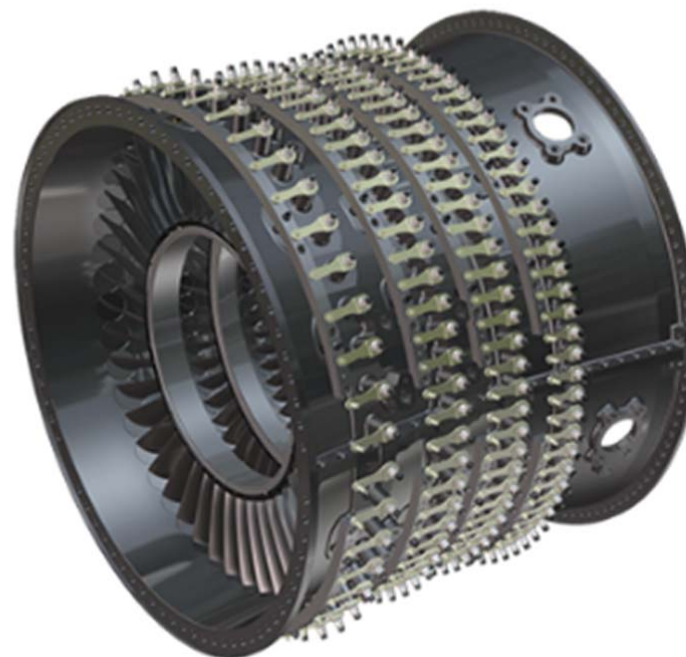


CORE ENGINE MAJOR MODULE – MM02 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM30 HPC Module ASSEMBLY



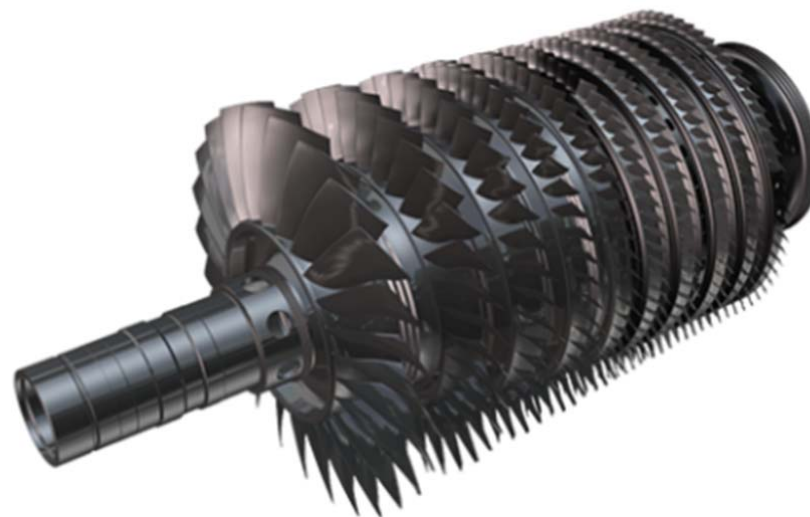


HPC ASSEMBLY – SM30 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM31– HPC Rotor assembly module





HPC ROTOR ASSEMBLY MODULE – SM31 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



**SM32 –HPC FWD Stator
Assembly module**





HPC FWD STATOR ASSEMBLY – SM32 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM41 – Combustor Diffuser Nozzle assembly module



COMBUSTOR DIFFUSER NOZZLE ASSEMBLY – SM41 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



**SM42 – Combustion
chamber assembly
module**





COMBUSTION CHAMBER ASSEMBLY – SM42 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM51 – Stg1 HPT nozzle assembly module





STG 1 HPT NOZZLE ASSEMBLY – SM51 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM52 – HPT Rotor assembly module





HPT ROTOR ASSEMBLY – SM52 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM53 - STAGE 2 HPT Nozzle module





STG 2 HPT NOZZLE – SM53 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM54 – Turbine Center Frame module





TURBINE CENTER FRAME – SM54 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM55 – N°5 Bearing Support



N°5 Bearing Support – SM55 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM56 – LPT Stage 1 nozzle



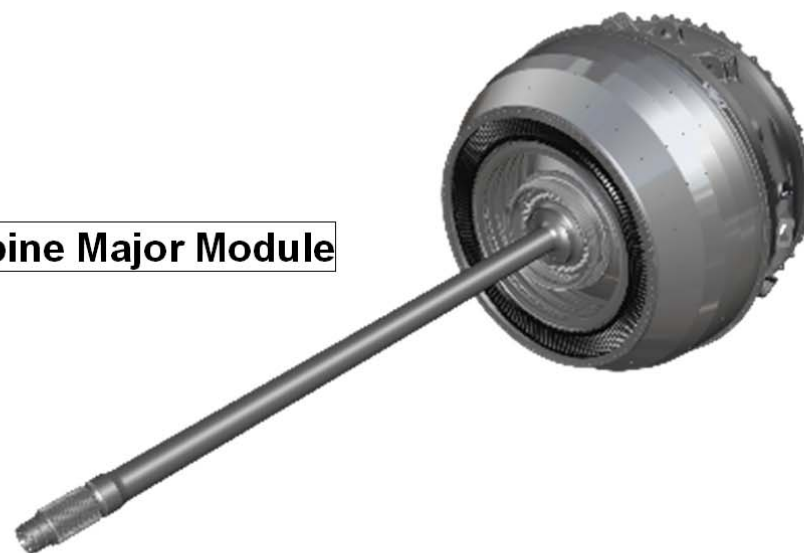


LPT STG 1 NOZZLE – SM56 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



MM3 : 72-00-03 - Low Pressure Turbine Major Module





LPT MAJOR MODULE – MM03 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM57 – LPT Shaft module





LPT SHAFT – SM57 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM58 LPT Rotor and Stator Module





LPT ROTOR & STATOR – SM58 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM59 TURBINE REAR VANE MODULE



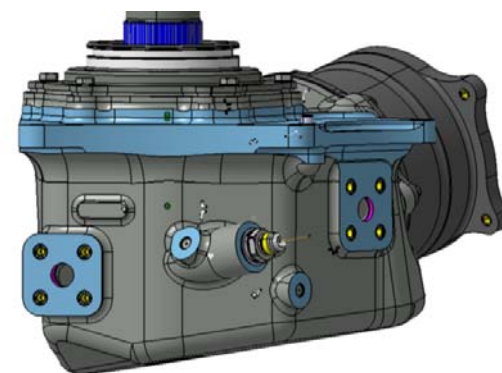
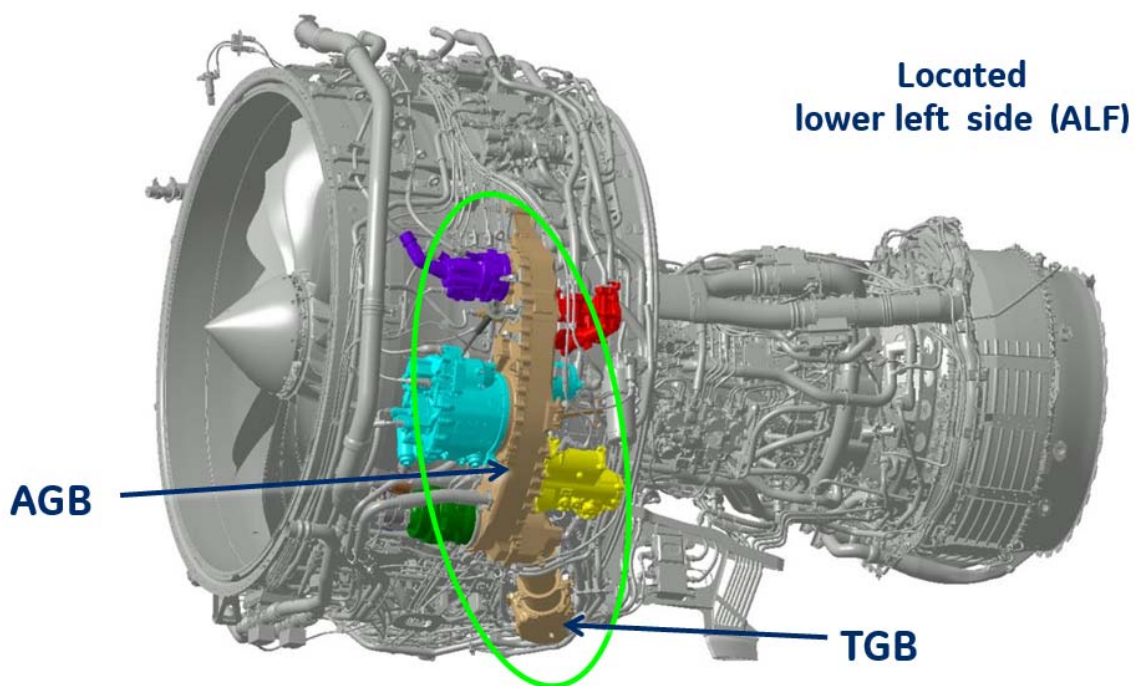


TURBINE REAR FRAME – SM59 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM62 – Transfer Gearbox (TGB) module





TRANSFER GEARBOX (TGB) – SM62 – Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SM63 – Accessory Gearbox (AGB) module



ACCESSORY GEARBOX (AGB) – SM63 – Recommended Service Bulletins

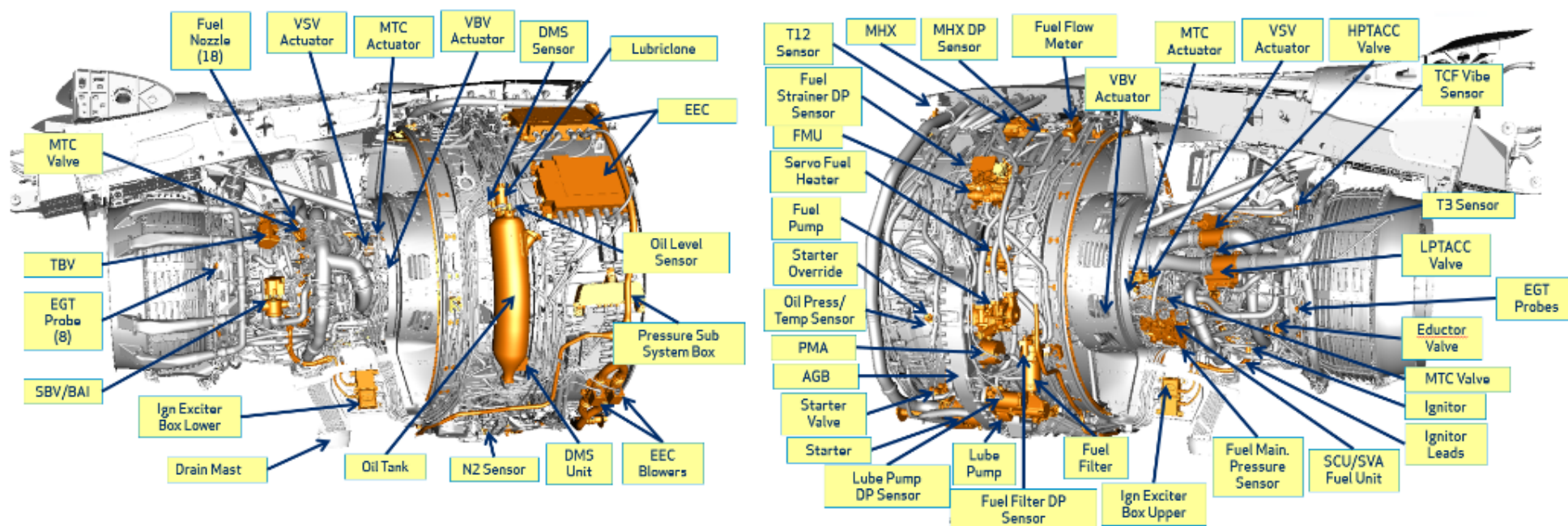


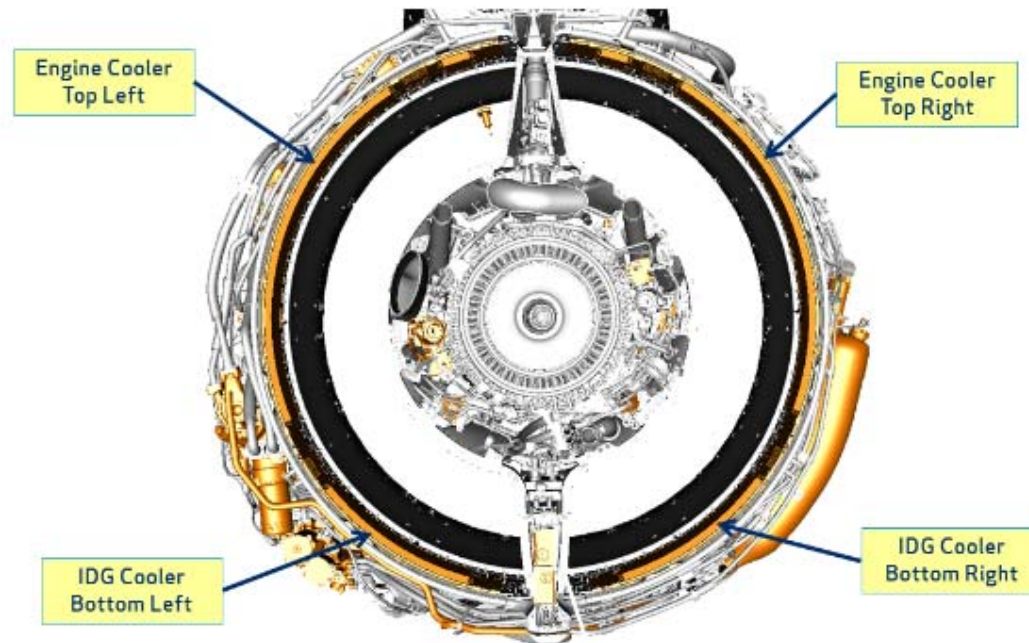
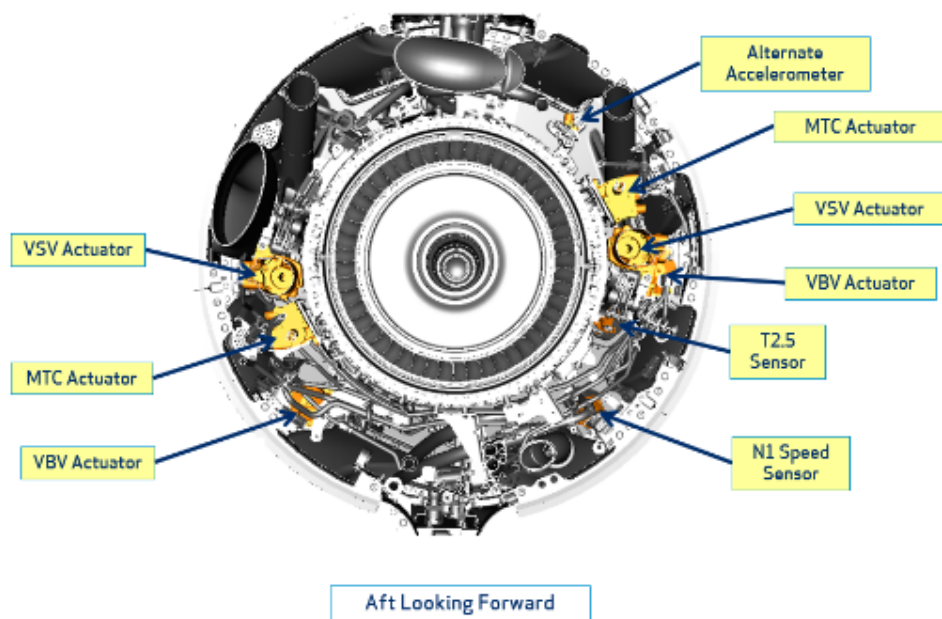
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Section 5 : Controls and Accessories Maintenance Guide









Workscope level definition for Controls & Accessories

Choice of maintenance level on C&A should be determined according to maintenance action to be performed and SB to be applied. Each SB needs appropriate level selection to be applied.

Level 00 Check Inspection

This level is applicable for Continued Time Engine Maintenance (CTEM), with C&A installed or removed for access.

- Perform check inspection of the control & accessory per ESM 72-00-00 \08 Check inspection.

Level 01 Minimum Workscope

This Level is applicable for non CTEM workscope with C&A installed or removed from the engine

Basically for this level of maintenance, The C&A is considered to be serviceable per on wing maintenance unless fault message detected and indication detected :

- Make sure that all engine related fault messages within the central maintenance computer (CMC) fault history or in the electronic engine control (EEC) and engine monitoring unit (EMU) non-volatile memory (NVM) have been reviewed and define appropriate level of maintenance if necessary.
- During visual examination at incoming inspection of the engine, if findings are noticed on installed C&A, refer to AMM criteria for serviceability
- If equipment is removed for access, perform visual examination and store properly the equipment. If findings Refer to AMM criteria for serviceability before reinstallation or storage.

This minimum Workscope may define some additive maintenance action (ex : filter replacement) or some S/B implementation that can be performed at this level of maintenance.

The individual C&A maintenance program will be updated following in service data and the Leading Indicator Program.

Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary

This level is applicable as soon as the C&A needs to be maintained per CMM procedure

The Control & Accessory should be cleaned, inspected and repaired or modified if required, without a full disassembly.
S/B implementation can lead to this level of maintenance

This level may be divided in sub levels per CMM if necessary to manage maintenance instructions

- **Level 02.01 Test / Fault Isolation or Visual examination**
It is recommended to do the CMM test and fault isolation procedures first to determine how much maintenance is necessary
- **Level 02.02 Partial Disassembly for SB implementation**
S/B incorporation per CMM can lead to this partial disassembly level.

Level 03 Overhaul per CMM



This level is applicable as soon as the Control & Accessory needs to be fully disassembled
S/B implementation can lead to this level of maintenance

Replace

The replacement of a specific control & accessory may be applied following Maintenance thresholds or inspected condition during Shop visit
S/B may ask for C&A replacement

Controls and accessories thresholds recommendation at engine shop visit

C&A maintenance is anticipated to be on-condition until maintenance threshold can be established if needed.

Maintenance thresholds are recommended maintenance time at shop visit above which specified levels of maintenance should be performed to maximize component reliability until next engine shop visit other than shop visit for convenience.

Maintenance thresholds have not been established and will be established if necessary using Lead Indicator Program results and in service data.
C&A Time since new and time since last shop maintenance (test or overhaul) must be recorded to ensure effective maintenance in case of a threshold recommendation is established.

Where there is a defined threshold, the Threshold counter is reset to zero when the C&A unit is Overhauled per the applicable CMM.



Engine C&A Workscopes Digest

Find following examples of Engine C&A workscopes for sheduled Shop visit.

Theses workscopes may be adapted to individual engine C&A Fault reports and will updated when Thereshold will be defined

Workscope A : Engine C&A minimum workscope

System	C&A Description	ATA	Removal status	Workscope level	Remarks or S/B to be applied
Fuel	Main Fuel Pump (MFP)	73-11-06		Level 1: Minimum workscope	
	Main Fuel Filter (MFF)	73-11-10		Level 1: Minimum workscope	
	Servo Fuel Heater (SFH)	73-11-25		Level 1: Minimum workscope	
	Fuel Nozzle	73-11-30		Level 1: Minimum workscope	
	A/C generator Air/Oil Cooler (IDG Cooler) (BOEING)	73-11-50		Level 1: Minimum workscope	
	Fuel Metering Unit (FMU)	73-21-06		Level 1: Minimum workscope	
	Split Unit Control/Servo Valve Assy (SCU/SVA)	73-21-10		Level 1: Minimum workscope	
	T12 Sensor	73-21-12		Level 1: Minimum workscope	
	T25 sensor	73-21-15		Level 1: Minimum workscope	
	T3 sensor	73-21-27		Level 1: Minimum workscope	
	Fuel Manifold Pressure (FMP) Sensor	73-21-36		Level 1: Minimum workscope	
	Permanent Magnet Alternator (PMA)	73-21-40		Level 1: Minimum workscope	
	Pressure Sub-system Box (PSS) and sensor	73-21-46		Level 1: Minimum workscope	
	FADEC (2 boxes (1/channel) - Electronic Engine Control (EEC)	73-21-56		Level 1: Minimum workscope	
	Data Entry Plug (old rating plug)	73-21-58		Level 1: Minimum workscope	
	Fuel Flow Meter (FFM)	73-31-05		Level 1: Minimum workscope	
	Fuel Filter Delta-P Sensor (FFDP),	73-34-05		Level 1: Minimum workscope	



System	C&A Description	ATA	Removal status	Workscope level	Remarks or S/B to be applied
	Fuel Strainer Delta-P Sensor (FSDP)	73-34-05		Level 1: Minimum workscope	
	Fuel/Oil Main Heat Exchanger Delta-P Sensor (MHXDP)	73-34-05		Level 1: Minimum workscope	
Ignition	Ignition Exciter box	74-11-07		Level 1: Minimum workscope	
	Igniter Lead	74-21-07		Level 1: Minimum workscope	Shop visit before 6000 hrs TOW Shop visit after 6000 hrs TOW Replace
	Igniter	74-21-10		Level 1: Minimum workscope	Shop visit before 1000 cycles TOW visually inspect per CMM after 1000 hrs TOW Replace
Air	Start Bleed / Booster Anti-Ice –SBV/BAI	75-11-05		Level 1: Minimum workscope	
	HPTACC Valve Assy/ Actuator	75-21-06		Level 1: Minimum workscope	
	LPTACC Valve/ Actuator	75-21-06		Level 1: Minimum workscope	
	Educator valve	75-21-45		Level 1: Minimum workscope	
	MTC Valve Actuator (without linkage and valve body) (set of 2)	75-23-05		Level 1: Minimum workscope	
	MTC Valve and Body (set of 2 per engine)	75-23-10		Level 1: Minimum workscope	
	FADEC BLOWER (2 per engine)	75-24-06		Level 1: Minimum workscope	
	VBV Actuator (Set of 2) Boeing	75-31-05		Level 1: Minimum workscope	
	VSV Actuator (set of 2)	75-32-05		Level 1: Minimum workscope	
	Transient Bleed Valve / Actuator	75-33-05		Level 1: Minimum workscope	
Indication	N1 Speed sensor	77-11-06		Level 1: Minimum workscope	
	N2 Speed sensor	77-11-11		Level 1: Minimum workscope	
	EGT Harness	77-21-06		Level 1: Minimum workscope	
	Sensor, T48 EGT	77-21-11		Level 1: Minimum workscope	
	Sensor, TCF or TRF vibe (accelerometer)	77-31-06		Level 1: Minimum workscope	



System	C&A Description	ATA	Removal status	Workscope level	Remarks or S/B to be applied
	Sensor, #1 brg vive or FF vibe (accelerometer)	77-31-11		Level 1: Minimum workscope	
Oil	Oil Tank	79-11-06		Level 1: Minimum workscope	
	Lubrication Unit	79-21-06		Level 1: Minimum workscope	
	Main fuel/oil heat exchanger (MHX)	79-21-20		Level 1: Minimum workscope	
	Engine ACOC (BOEING)	79-21-30		Level 1: Minimum workscope	
	Eductor Valve (using Ni Alloy version)	79-21-45		Level 1: Minimum workscope	
	Non Return Valve (NRV)	79-21-50		Level 1: Minimum workscope	
	Oil Level Sensor (OLS)	79-31-06		Level 1: Minimum workscope	
	Oil quantity/level transmitter/Sensor (OLS) (BOEING)	79-31-07		Level 1: Minimum workscope	
	Oil Pressure Temperature Transducer Sensor (OPTS)	79-32-05		Level 1: Minimum workscope	
	Oil Filter Delta Pressure Transducer (OFDP)	79-35-05		Level 1: Minimum workscope	
	AIR.OIL Separator (Lubriclone)	79-36-05		Level 1: Minimum workscope	
	ODMS Sensor	79-36-10		Level 1: Minimum workscope	
	ODM (Oil Debris Monitoring) harness	79-36-15		Level 1: Minimum workscope	
	Signal Conditioning ODM Unit - Oil debris monitoring unit	79-36-20		Level 1: Minimum workscope	
Starting	Pneumatic Air Starter	80-11-05		Level 1: Minimum workscope	
	Starter Air Valve	80-12-15		Level 1: Minimum workscope	
Harnesses	Engine Wiring Harness HJ3A	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ3B	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ5A	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ5B	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ6A	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ6B	73-21-60		Level 1: Minimum workscope	



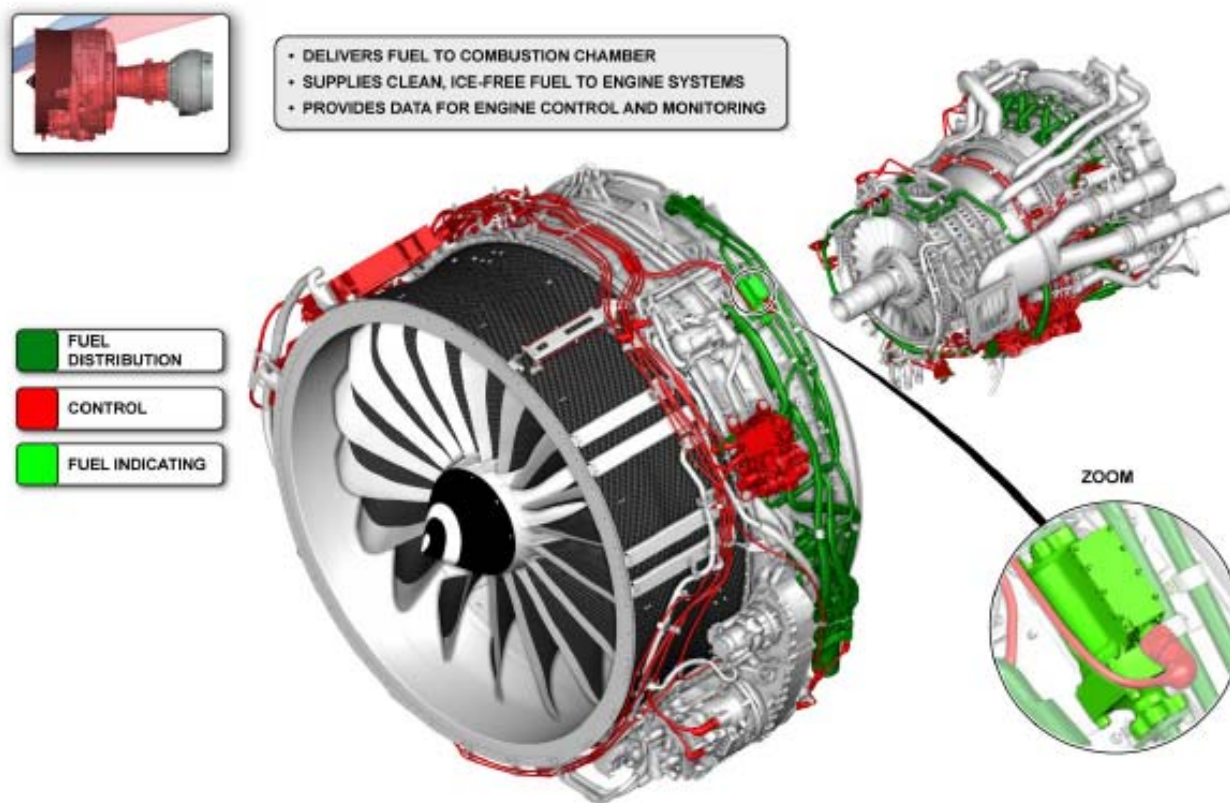
System	C&A Description	ATA	Removal status	Workscope level	Remarks or S/B to be applied
	Engine Wiring Harness HJ7	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ8A	73-21-60		Level 1: Minimum workscope	
	Engine Wiring Harness HJ8B	73-21-60		Level 1: Minimum workscope	

Workscope B : Engine C&A Test workscope

To be defined



Fuel system - ATA 73



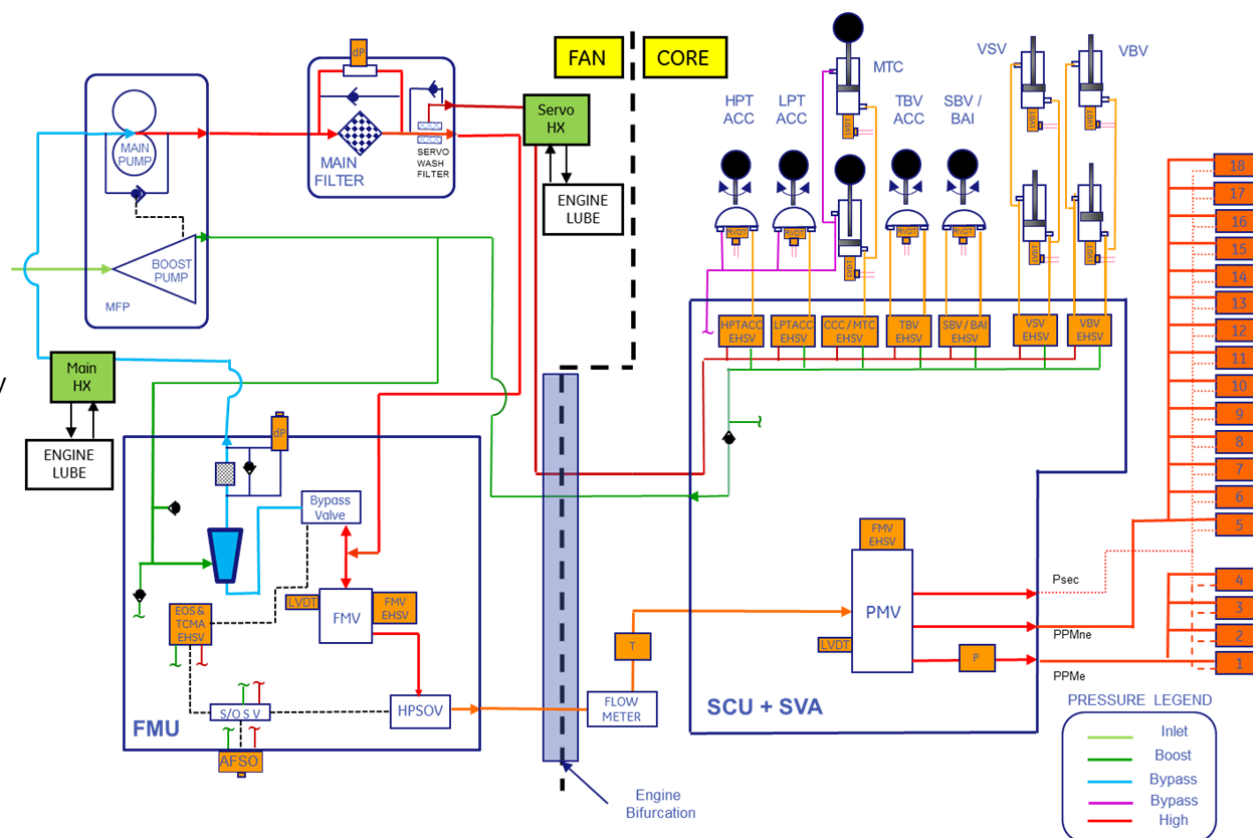
Engine Fuel and Control



Fuel System Overview

Key components

- Main Fuel Pump (MFP)
- Fuel Metering Unit (FMU)
- Fuel Filter Delta-P Sensor (FFDP),
- Fuel Strainer Delta-P Sensor (FSDP)
- MHX Delta-P Sensor
- Main Fuel Filter (MFF)
- Fuel Flow Meter (FFM)
- Split Control Unit/Servo Valve Assembly (SCU/SVA)
- Fuel Manifold Pressure (FMP) Sensor
- Fuel Nozzles





MAIN FUEL PUMP (MFP) WORKSCOPE

Workscope management



Accessory	Workscope level designation	Recommendation or Soft time thresholds
Main Fuel Pump (MFP) ATA 73-11-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



SERVO FUEL HEATER (SFH) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Servo Fuel Heater (SFH) ATA 73-11-25	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL METERING UNIT CONTROL (FMU) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Metering Unit Control (FMU) ATA 73-21-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Split Control Unit/Servo Valve Assembly (SCU/SVA) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Splitter Control Unit Control Assembly ATA 73-21-10	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



MAIN FUEL FILTER (MFF) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Main Fuel Filter ATA 73-11-10	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	Perform a Partial or Full overhaul per CMM instructions



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL MANIFOLD PRESSURE (FMP) SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Manifold Pressure Sensor ATA 73-21-36	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

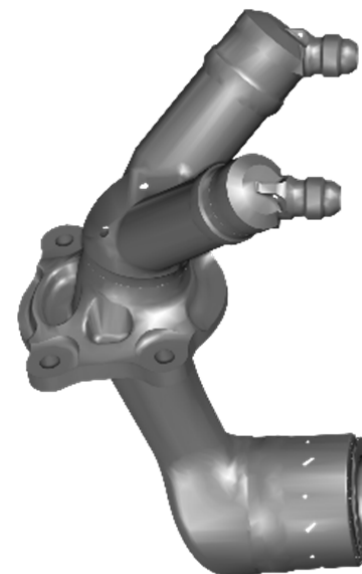
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL NOZZLE WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Nozzle ATA 73-11-30	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



PERMANENT MAGNET ALTERNATOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Permanent Magnet Alternator ATA 73-21-40	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

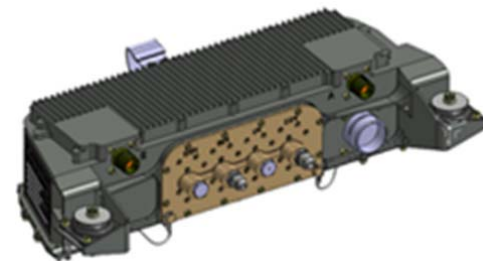
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



PRESSURE SUB-SYSTEM (PSS) BOX WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Pressure Sub-System (PSS) Box ATA 73-21-46	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

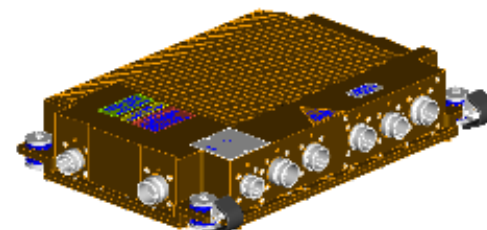
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



ELECTRONIC ENGINE CONTROL (EEC) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Electronic Engine Control (EEC) ATA 73-21-56	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL FLOW METER (FFM) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Flow Meter (FFM) ATA 73-31-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



RATING PLUG WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Data Entry Plug ATA 73-21-58	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL DIFFERENTIAL PRESSURE TRANSDUCER WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Differential Pressure Transducer ATA 73-34-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



FUEL STRAINER DELTA PRESSURE SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Fuel Strainer Delta Pressure Sensor ATA 73-34-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

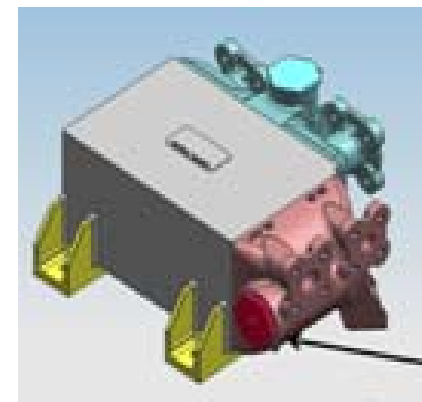
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

A/C GENERATOR AIR/OIL COOLER (IDG cooler) WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
A/C generator Air/Oil Cooler (IDG Cooler) ATA 73-11-50	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

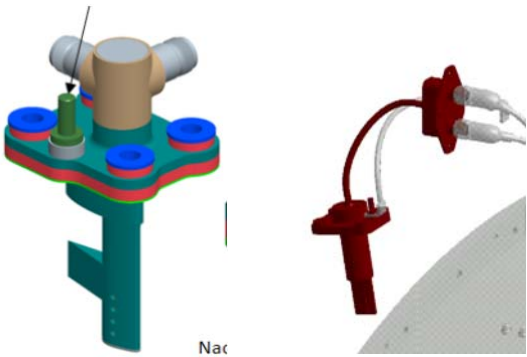
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

T12 SENSOR WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
T12 Sensor ATA 73-21-12	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

T25 SENSOR WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
T25 Sensor ATA 73-21-15	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

T3 SENSOR WORKSCOPE





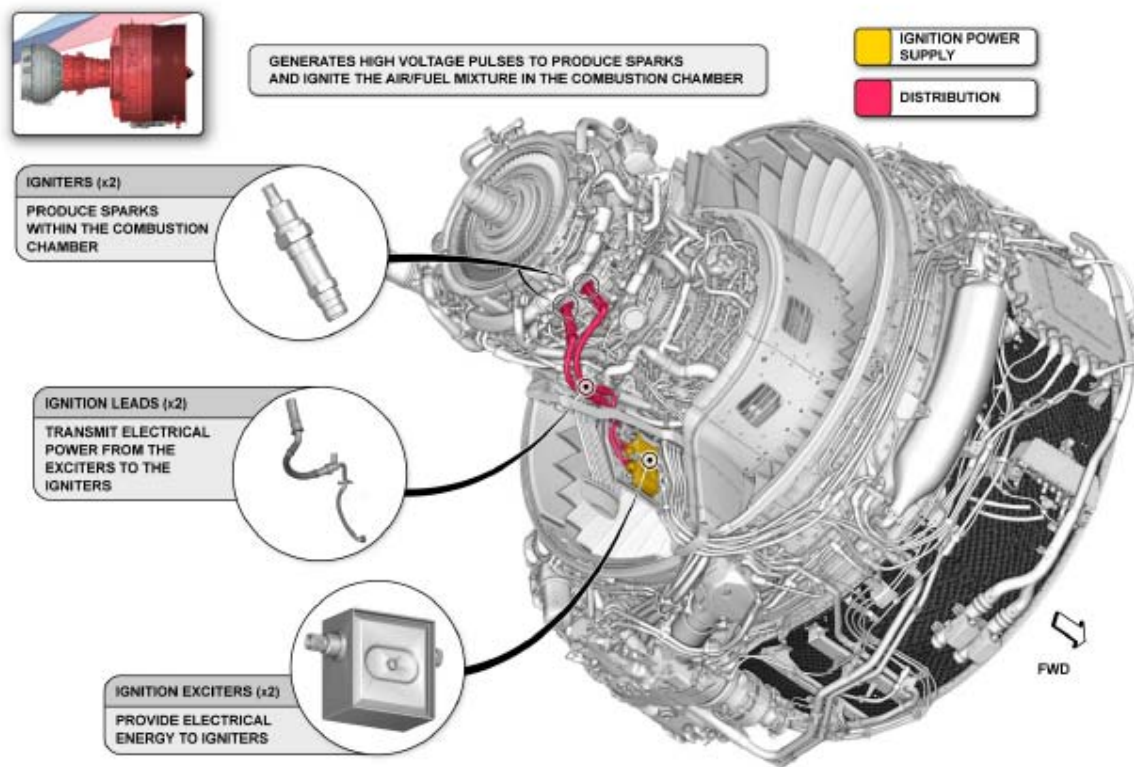
Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
T3 Sensor ATA 73-21-27	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

Ignition system - ATA 74



Ignition Components



IGNITION LEAD RIGHT AND LEFT WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Ignition Lead ATA 74-21-07	Level 00 Check inspection	Detailed visual inspection per AMM
	Level 01 Minimum WS	Shop visit before 6000 hrs TOW
	Replace	Shop visit after 6000 hrs TOW Replace



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



IGNITION EXCITER BOX RIGHT AND LEFT WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Ignition Exciter Left ATA 74-11-07	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

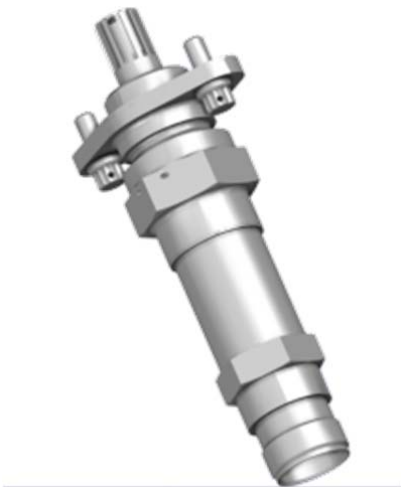
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



IGNITER RIGHT AND LEFT WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Left Igniter ATA 74-21-10	Level 00 Check inspection	
	Level 01 Minimum WS	Shop visit before 1000 cycles TOW visually inspect per CMM
	Replace	Shop visit after 1000 cycles TOW



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

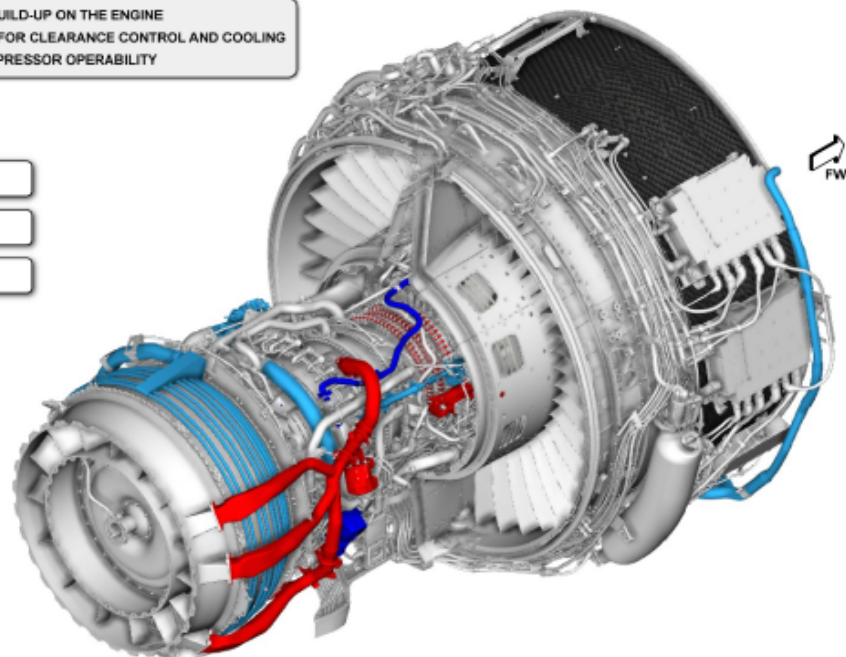
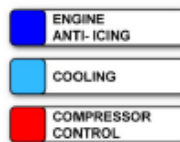


Air system - ATA 75

AIR SYSTEM OVERVIEW

Electronic Engine Control (EEC) Cooling
Variable Stator Vanes (VSVs)
Variable Bleed Valves (VBVs)
High Pressure Turbine Active Clearance Control (HPTACC)
Low Pressure Turbine Active Clearance Control (LPTACC)
Modulated Turbine Cooling (MTC)
Start Bleed Valve/Booster Anti-Icing (SBV/BAI)
Transient Bleed Valve (TBV)
Eductor Valve

- PREVENTS ICE BUILD-UP ON THE ENGINE
- REGULATES AIR FOR CLEARANCE CONTROL AND COOLING
- CONTROLS COMPRESSOR OPERABILITY



Air System



EEC COOLING BLOWERS WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
EEC Cooling Blowers ATA 75-24-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



VARIABLE STATOR VANE (VSV) ACTUATOR WORKSCOPE

Workscope management

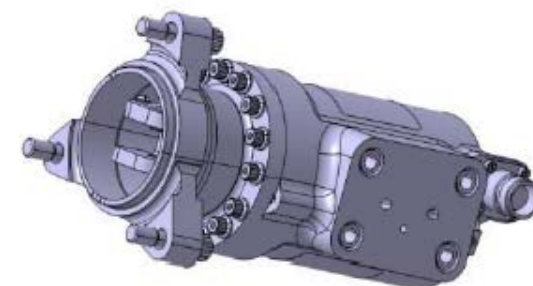
Accessory	Workscope level designation	Recommendation or Soft time thresholds
Variable Stator Vane (VSV) Actuator Set of 2 ATA 75-32-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

VBV ACTUATOR WORKSCOPE





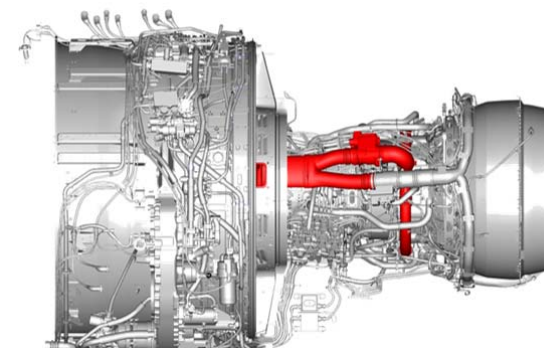
Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
VBV Actuator Set of 2 ATA 75-31-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

HPTACC VALVE WORKSCOPE





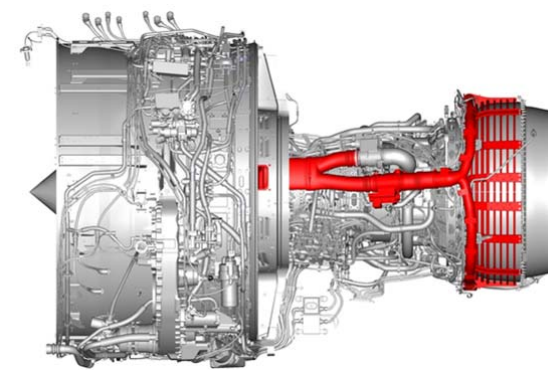
Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
HPTACC Valve ATA 75-21-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

LPTACC VALVE WORKSCOPE





Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
LPTACC Valve ATA 75-22-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

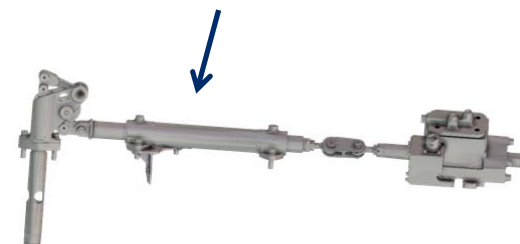
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

MTC VALVE & LINKAGE WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Modulated Turbine Cooling (MTC) Valve and Linkage ATA 75-23-10	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

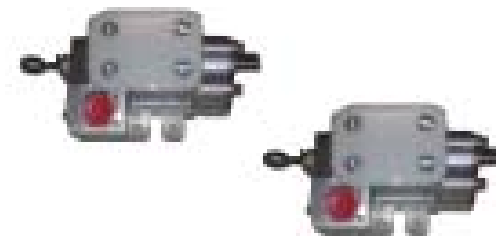
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



MTC ACTUATOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Modulated Turbine Cooling (MTC) Actuator ATA 75-23-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

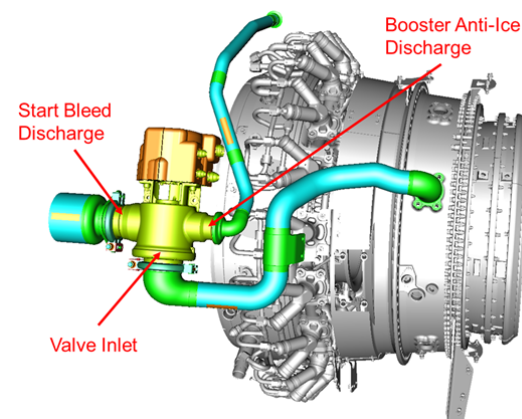
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



START BLEED/ BOOSTER ANTI-ICE –SBV/BAI WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Start Bleed/ Booster Anti-Ice –SBV/BAI ATA 75-11-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



TRANSCIENT BLEED VALVE (TBV) AIR VALVE WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Transcient Bleed Valve (TBV) Air Valve ATA 75-33-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

EDUCOR VALVE WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Educor Valve ATA 75-21-45	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

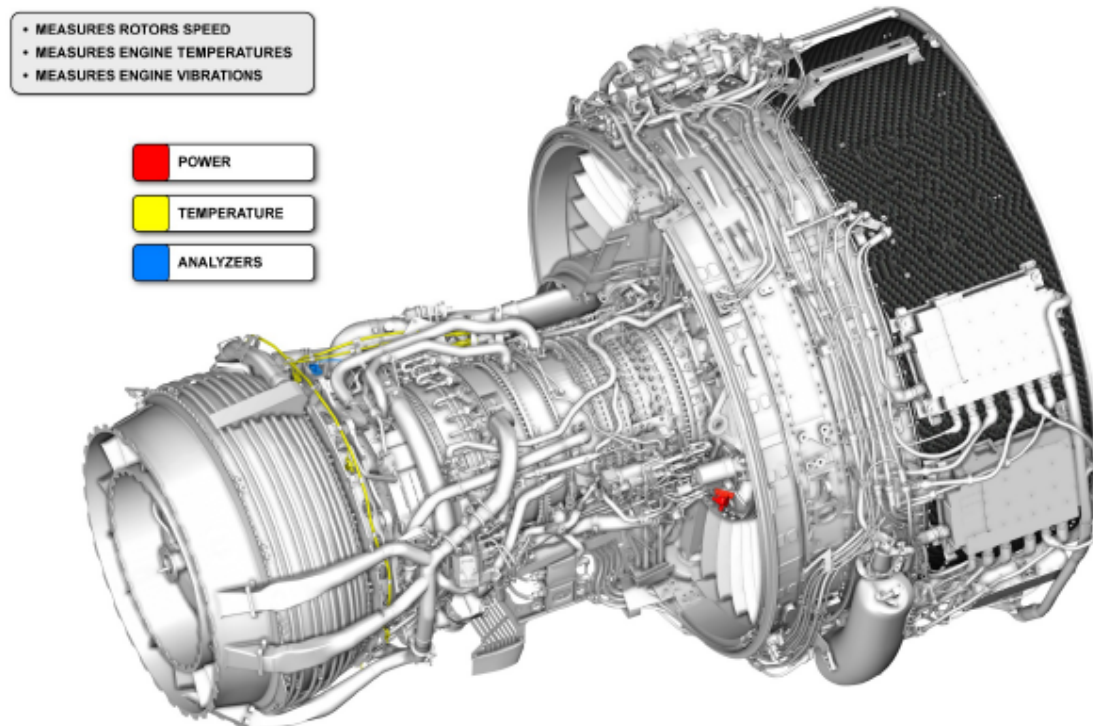


Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Indication system - ATA 77



Indicating Systems



N1 SPEED SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
N1 Speed Sensor ATA 77-11-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



N2 SPEED SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
N2 Speed Sensor ATA 77-11-11	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



T48 EGT SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
T48 Exhaust Gas Temperature Sensor (T48) ATA 77-21-11	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

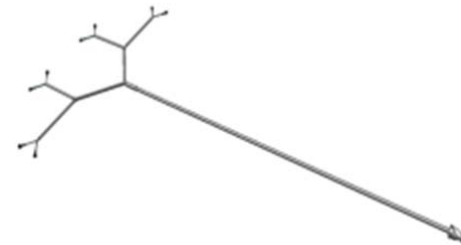
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



T48 EGT HARNESS WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
T48 Exhaust Gas Temperature (EGT) Harness Right and Left ATA 77-21-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



TCF VIBRATION SENSOR WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Turbine Centrale Frame Vibration Sensor ATA 77-31-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



#1 BRG VIBE SENSOR WORKSCOPE

Workscope management

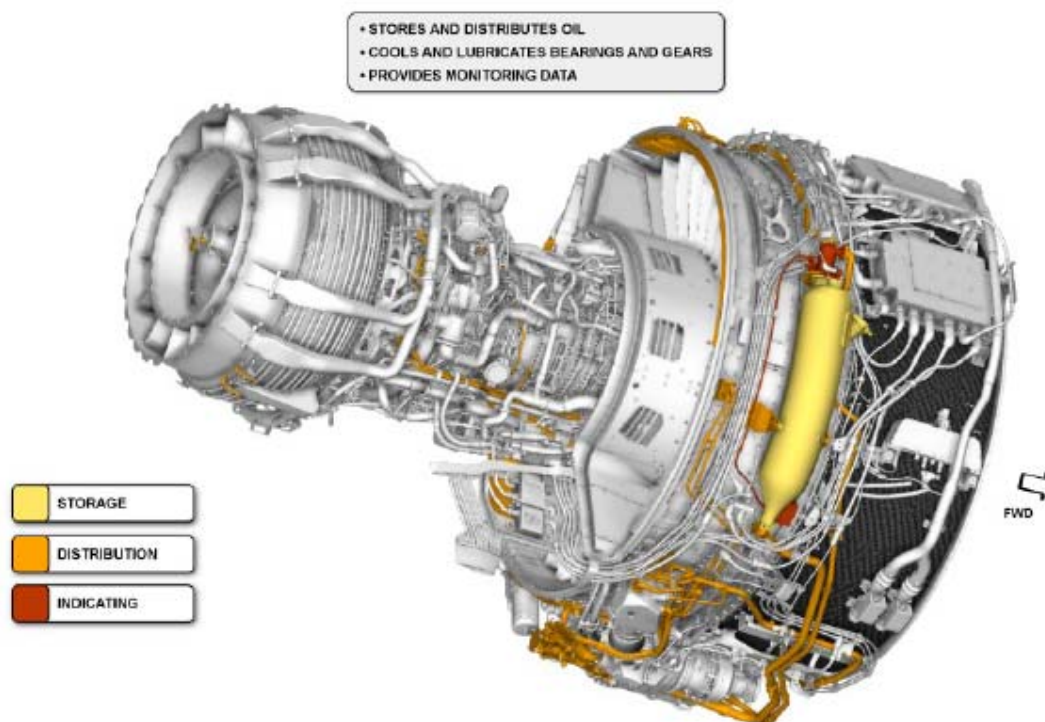
Accessory	Workscope level designation	Recommendation or Soft time thresholds
#1 BRG VIBE SENSOR ATA 77-31-11	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Oil system - ATA 79



Oil System



Lubrication System overview

Key elements:

Oil Tank

Lubrication Unit

Servo Fuel Heater (SFH)

Surface Air Cooled Oil Cooler (SACOC)

Main Heat exchanger (MHX)

5 main bearings located in 3 different engine sump

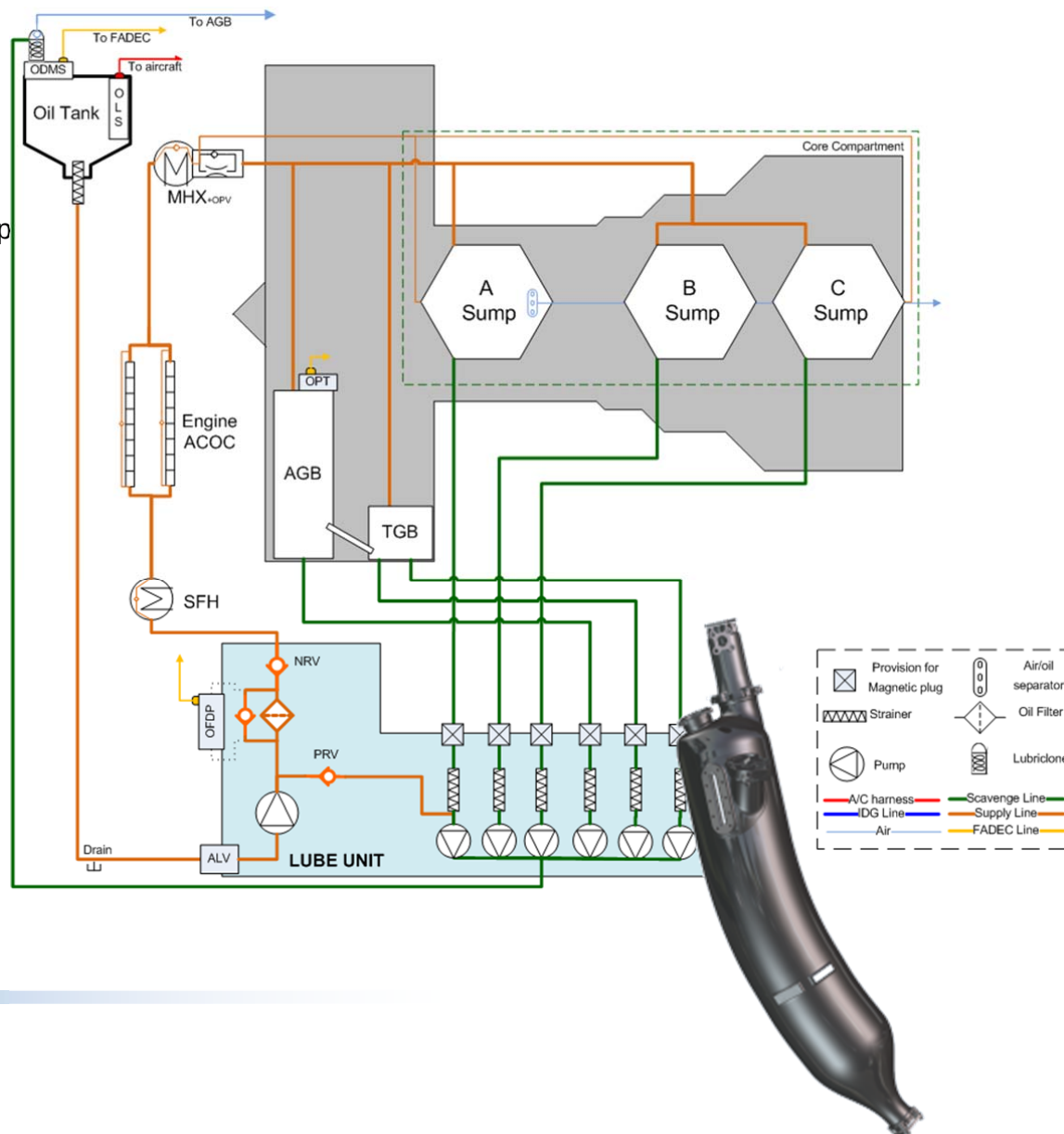
Monitoring:

Oil Filter DP

Oil Pressure & Temperature

Oil Debris

OIL TANK WORKSCOPE





Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Tank ATA 79-11-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

LUBRICATION UNIT WORKSCOPE





Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Lubrication Unit ATA 79-21-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

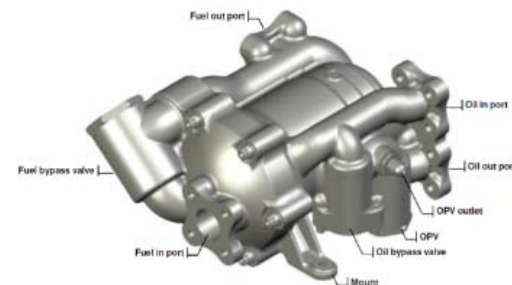
S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

MAIN HEAT EXCHANGER (MHX) WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Main Heat Exchanger (MHX) ATA 79-21-20	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

Engine ACOC WORKSCOPE





Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Engine Air Cooling Oil Cooler (ACOC) ATA 79-21-30	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

ODM HARNESS WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Debris Monitoring harness (ODMS) ATA 79-36-15	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

OIL DEBRIS MONITORING UNIT WORKSCOPE



Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Debris Monitoring Monitoring Unit ATA 79-36-20	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



OIL LEVEL SENSOR (OLS) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Level Sensor (OLS) ATA 79-31-06	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



OIL PRESSURE / TEMPERATURE TRANSDUCER SENSOR (OPTS) WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Pressure Temperature Transducer Sensor (OPTS) ATA 79-32-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



OIL FILTER DELTA PRESSURE TRANSDUCER (OFDP) –WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Oil Filter Delta Pressure Transducer (OFDP) ATA 79-35-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	



Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



EDUCTOR VALVE WORKSCOPE

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Eductor Valve ATA 79-21-45	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

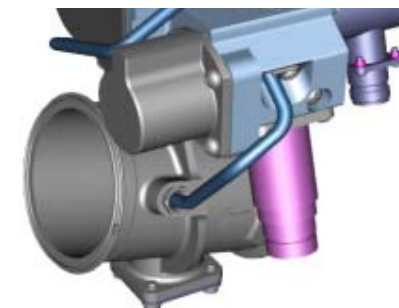


Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

Starting system - ATA 80

STARTER AIR VALVE WORKSCOPE





Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Starter Air Valve ATA 80-12-15	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

PNEUMATIC AIR STARTER WORKSCOPE



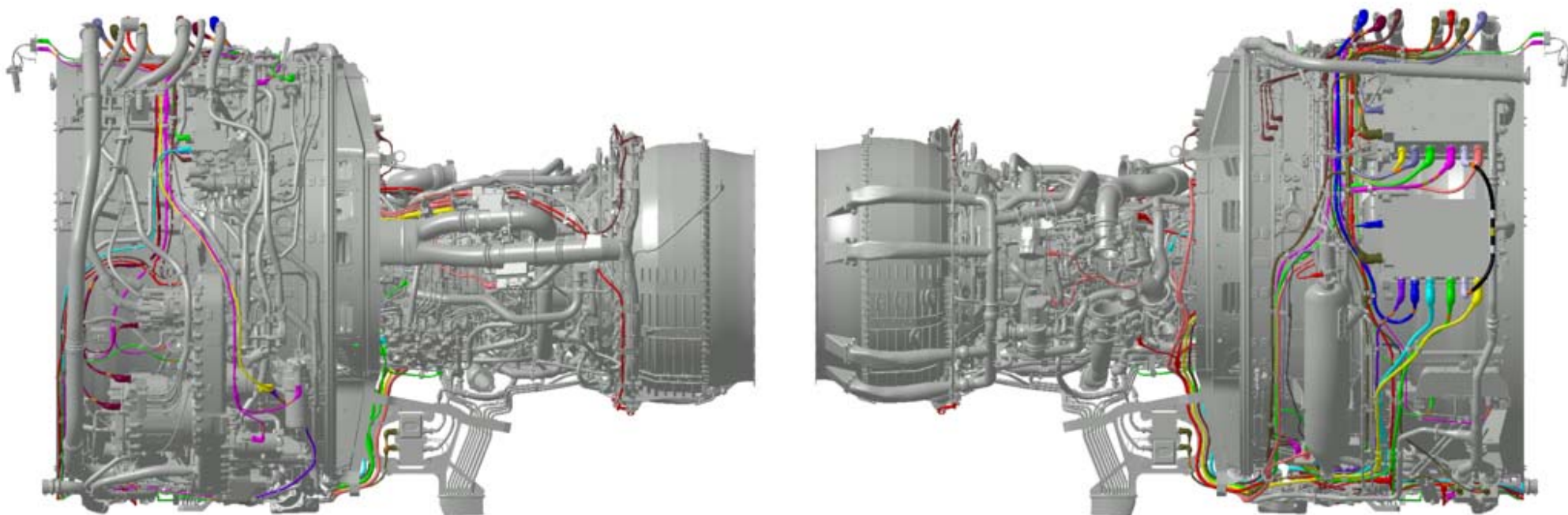
Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
Pneumatic Air Starter ATA 80-11-05	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

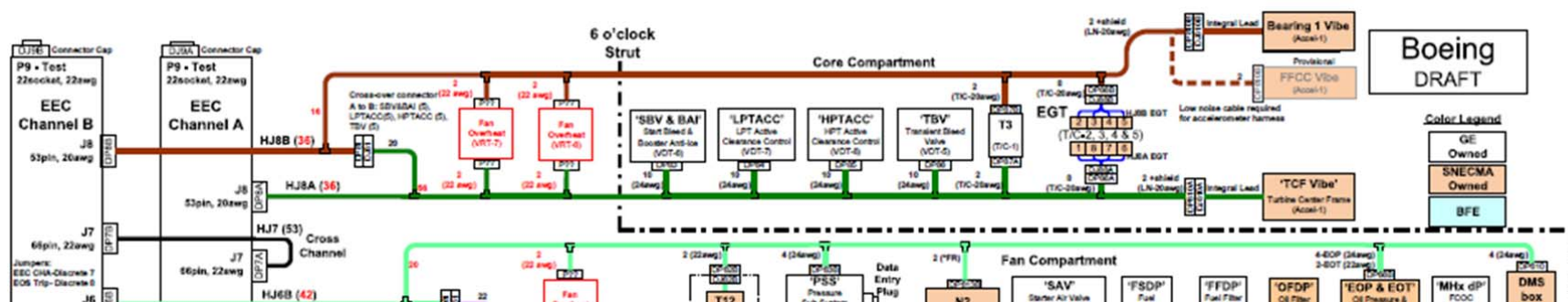
Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation

Harnesses



Harness Electrical diagram







HARNESSES : HJ3A, HJ3B, HJ5A, HJ5B, HJ6A, HJ6B, HJ7, HJ8A, HJ8B

Workscope management

Accessory	Workscope level designation	Recommendation or Soft time thresholds
HARNESSES : HJ3A,HJ3B, HJ5A,HJ5B, HJ6A, HJ6B, HJ7, HJ8A, HJ8B ATA 73-21-60	Level 00 Check inspection	
	Level 01 Minimum WS	
	Level 02 Inspection / Test & Fault Isolation per CMM and Repair as necessary	
	Level 03 Overhaul per CMM	

Recommended Service Bulletins

S/B #	Document designation	Compliance	Impact statements	S/B Description and PSE recommendation



Suppliers & Content

Kulite

Fuel Manifold P Sensor
Fuel Filter dP Sensor
Fuel Strainer dP Sensor

Ametek

Oil Filter dP Sensor
Oil P & T Sensor

Unison

IDG SACOC
Engine Surface Air Cooling Oil Cooler
Fuel temperature resistance detector
Rating plug
Alternator Stator
Alternator Rotor
Fuel Temp Sensor
T3 Sensor
Core Compartment Temp Sensor
Ignition Exciter
Ignition Lead
Igniter

FADEC Alliance

Pressure Subsystem
EEC

Crane

Fuel Flow Transmitter



Honeywell

Eductor Valve
Ejector Valve
VBV Actuator
Starter Air Valve
Pneumatic Air starter
Pneumatic Air starter magnetic plug
Manual Override Assy

Woodward

FMU
SVA
Fuel Filter Module
Fuel Pump
MTC Valve Actuator
HPTACC Valve
LPTACC Valve
Transient Bleed Valve
VSV Actuator
SBV Valve Assy

Meggitt

MTC Valve
EGT Harness
Servo Fuel Heater
N1 Sensor
N2 Sensor
T12 Sensor
T25 Sensor
EGT Sensor
Vibe Sensors
Oil Level Sensor
Main Fuel/Oil HX



Workscopes Access to Equipments : Engine Kit Overview

